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Review

Intermediate phase engineering of halide perovskites for photovoltaics

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SUMMARY

Metal halide perovskites serving as one of the most promising photovoltaic materials are gaining considerable attention worldwide. For achieving high performance as well as long-term stability of the perovskite solar cells (PSCs), a good quality of perovskite film featuring in smooth, pinhole-free morphology, full coverage over substrates, good heterojunction contacts, and stable photoactive phase is of great importance. During solution fabrication for perovskite films, intermediate phase, which refers to the state of precursor composition before final annealing, plays an essential role in determining the film quality, especially in the state-of-the-art PSCs. In this review, we summarize the research involving mechanism and applications of intermediate phase engineering (IPE) processes in various solution-processing technologies. The challenges of IPE are further discussed, and perspectives are provided for developing high-performance PSCs via IPE.

INTRODUCTION

Metal halide perovskite materials with outstanding optoelectronic properties such as long carrier diffusion length, tunable band gaps and high defect tolerance are gaining worldwide attention in photovoltaics,[1–3] light-emitting diodes,[4,5] photodetectors,[6,7] etc. In photovoltaics, the certified power conversion efficiency (PCE) of a single-junction perovskite solar cell (PSC) has reached 25.7% in 2021[8] since PSCs were first reported in 2009.[9] The PCEs of a minimodule (63.98 cm^2) and a small module (804 cm^2) devices have also reached 20.1%[10] and 17.9%,[11] respectively. In addition, the operational lifetime of PSCs has achieved ~1 year under controlled standard aging conditions.[12] Moreover, variable band gaps and easy film-processing techniques make perovskites compatible with multiple photo absorbers, such as silicon (Si), copper indium gallium selenide (CIGS), organic, and quantum dots, for fabricating multijunctional solar cells. The certified PCE has reached 29.8% for perovskite/Si,[13,8] 24.2% for perovskite/CIGS,[14] and over 26% for all-perovskite tandem solar cells.[10]

Perovskite has a typical lattice formula of ABX_3, where A is a monovalent cation (such as methylammonium [MA^+], formamidinium [FA^+], and Cs^+), B is a divalent metal cation (such as Pb^{2+}, Sn^{2+}, and Ge^{2+}), and X-site is occupied by halide anions (such as I^-, Br^-, and Cl^-). The crystal structure of ABX_3 perovskite features the corner-sharing $[BX_6]^{4-}$ octahedra, where A cations fill the space in between. Such structure adopts a cubic phase for a small band gap and dispersive bands for low effective masses of charges.[15]

Context & scale

Metal halide perovskites have sparked immerse research interests for highly efficient photovoltaic applications owing to their outstanding photophysical properties and scalable solution fabrication processes. Among the state-of-the-art solution-processed perovskite solar cells (PSCs), intermediate phase engineering (IPE), characterized by precise manipulation or control of precursor states before annealing, is playing an increasingly important role in forming high-quality perovskite film, which is essential to high-performance cells with long-term stability. In this review, we summarize the roles and mechanisms of IPE in different film-processing technologies, including two-step method, one-step solution method, methylamine gas post-healing or solvation process, antisolvent method, and volatile additives method. The challenges of IPE are discussed, and perspectives are provided for the developing high-performance PSCs via IPE.

Low nucleation and crystallization activation energies of halide perovskites (56.6–97.3 kJ mol^{-1})[16] allow simple solution processing of perovskite films by spin-coating in laboratories, as well as large-scale fabrication with slot-die coating, ink-jet printing, roll-to-roll printing for industrial applications. The solution-processing techniques usually include precursor solution process, intermediate phase process (including coating process and initially deposited films), and crystal growth process during film annealing (Figure 1A). In perovskite precursor solutions, polar aprotic solvents, such as dimethylformamide (DMF) and dimethyl sulfoxide (DMSO), are commonly employed due to their strong coordination abilities with precursors.[17–19] During the formation of fresh films from e.g., $MAPbI_3$ precursor solutions, intermediate phases in the precursor film are generated, such as MAI-PbI_2-DMF and MAI-PbI_2-DMSO. These perovskite intermediate phases incline to grow in the direction of (020) plane from polar solvents at room temperature (RT), leading to the needle-like perovskite films with some pinholes (Figure 1B).[20] In order to prepare uniform and pinhole-free perovskite films, disturbing the crystal growth kinetics or changing the component of intermediate phase is necessary (Figure 1C).[21]

The solution-processed antisolvent method is the state-of-the-art deposition technology for high-efficiency PSCs.[23] The drop of antisolvent onto the precursor films can quickly extract the residual solvent, inducing fast supersaturation and leading to a large amount of crystal nucleus within a short period of time. It avoids the further growth of the intermediates in the direction of (020) plane and results in the generation of dense perovskite films. In addition, adjusting the composition of intermediates in the fresh film can increase the grain size based on the antisolvent method.[24] Apart from one-step antisolvent method, two-step methods (such as dipping and two-step spin-coating) by sequentially depositing PbX_2 and AX precursors have also been widely studied, which modulates the process of crystallization to obtain uniform perovskite films.[25,26] To overcome the inadequate reaction of PbI_2 layer and MAI in two-step method, PbI_2-DMSO precursor films are first deposited to gain pure perovskite films through subsequent intramolecular exchange between MA^+ and DMSO.[27] The introduction of additives (such as MACl, MAAc, NH_4Cl, etc.) into the precursor solutions results in forming new intermediates and changing the crystal growth process for desirable morphology of perovskite films.[2,28–34] The organic-inorganic hybrid perovskite (OIHP) intermediate phase can also improve the interfacial contact of inorganic perovskite films with metal oxide charge transport layers due to the strong interaction between the two.[31] What is more, some volatile salt additives, such as MACl, FACl, DMAI (DMA: dimethylammonium), and MAI, have been introduced into $FAPbI_3$ or $CsPbI_3$ perovskite precursor solutions to promote the formation of the corresponding pure black perovskite phases, as the generation of these intermediates during film processing avoids the formation of the corresponding yellow phases.[35,36] Herein, the strategies of adjusting intermediate phases for the formation of high-quality perovskite films, good interfacial contact, and favored phase transformation are named as intermediate phase engineering (IPE).

Intermediate phase is formed by coordinating perovskite precursors with solvents and/or additives before the formation of final films and is closely related to the coordination ability of Pb^{2+} and the coordination compounds. Donor number (D_N) was defined by Gutmann to determine the coordination ability of the solvent with solute.[37] Solvents with larger D_N suggest stronger capability to coordinate with Pb^{2+} center, which leads to more stable intermediate phases (Figure 1D).[22] The addition of organic ammonium cations (AX) weakens the interaction between

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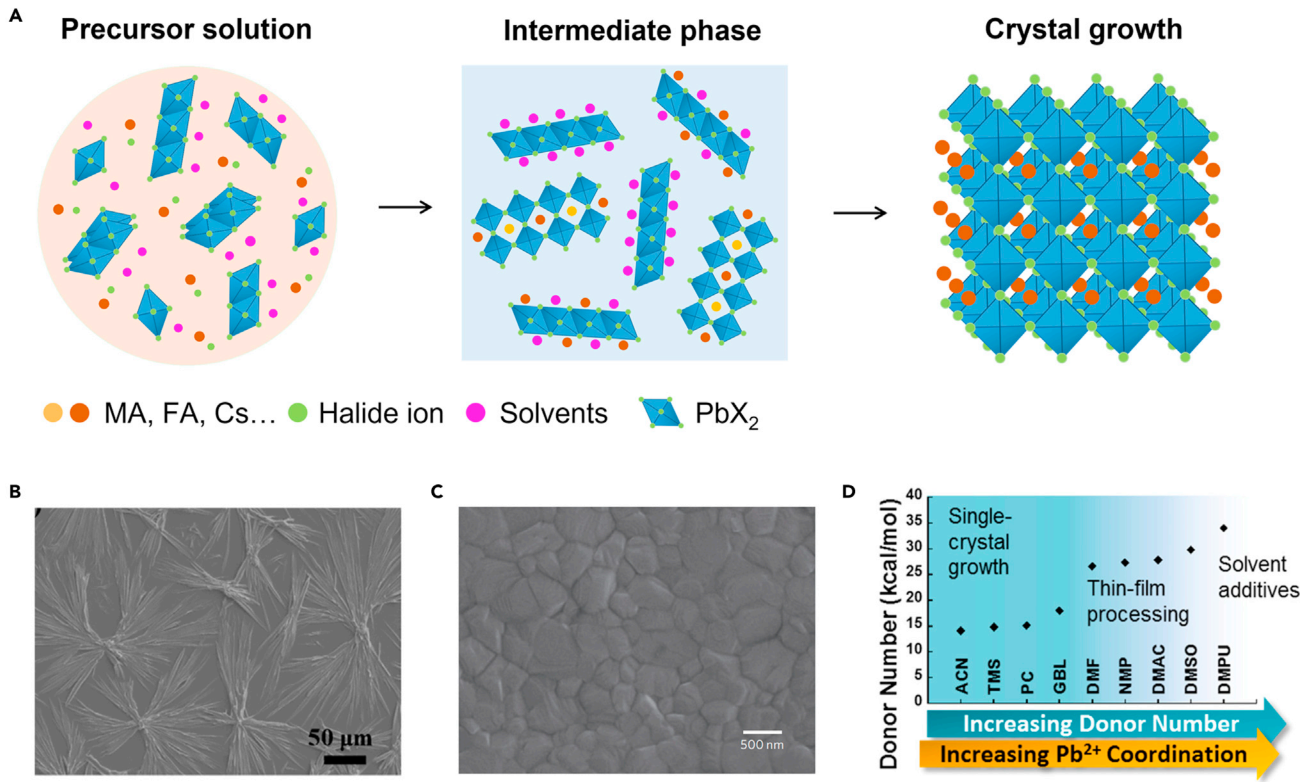

Figure 1. Formation of perovskite films via intermediate phase engineering (IPE) and solvent coordination

(A) Scheme of the formation of solution-processed perovskite films, including precursor solution process, intermediate phase process, and crystal growth process during film annealing.

(B) The scanning electron microscope (SEM) image of an $MAPbI_3$ intermediate film prepared from droplets of precursor solution with DMSO solvent at room temperature. Copyright from ACS Publications (2018).[20]

(C) The SEM image of a perovskite film with dense-grained uniform morphology with intermediate phase engineering in antisolvent method. Copyright from Nature Publishing Group (2014).[21]

(D) Gutmann's donor number (D_N) of solvents for perovskite film processing (ACN, acetonitrile; TMS, tetramethylsilane; PC, propylene carbonate; GBL, γ-butyrolactone; NMP, *N*-Methylpyrrolidone; DMAC, dimethylacetamide; DMPU, N,N′-dimethylpropyleneurea). Copyright from ACS Publications (2018).[22]

solvents and Pb^{2+} by intercalation into a closer position to the $[PbI_x]$ framework, leading to the formation of different intermediate phases.

In this review, we summarize the roles and mechanisms of IPE in different film-processing techniques including two-step method, one-step solution method, MA gas post-healing or solvation process, antisolvent method, and volatile additives method. As the PbX_2/AX ratio affects the species of intermediate phase, the IPE is classified into three types based on the coordination numbers of PbX_2 with AX (Table 1): (1) IPE based on PbX_2 and solvents (PbX_2-solvent), (2) IPE based on perovskite precursors and solvents (PbX_2-AX-solvent), and (3) IPE based on perovskite precursors and volatile additives (PbX_2-AX-*x*A′X′. A′X′ represents volatile additives). Based on the understandings of the fundamentals of IPE, we also provide perspectives on the development of high-performance PSCs with long-term operational stability.

IPE based on PbX_2 and solvents

Intermediate phases of PbX_2 and solvents are widely studied in two-step methods, which can promote full transformation of PbX_2 into perovskite phase by

Table 1. Roles of intermediate phases in various classifications of IPE and film-processing techniques

IPE classifications	Film-processing techniques	Roles of intermediate phases
PbX_2-solvent	two-step method	promoting the transformation of PbX_2 into perovskite phase
PbX_2-AX-solvent	one-step method	adjusting the crystallization process for uniform perovskite films
	MA gas post-healing or solvation process	liquid intermediate phase $MAPbI_3$-xMA promoting the formation of uniform perovskite films
	antisolvent method	prolonging the processing window of antisolvent dropping for uniform perovskite films
PbX_2-AX-A'X'	volatile additives	changing the perovskite crystal growth process for uniform perovskite films
		formation of organic-inorganic hybrid perovskite intermediate phase promoting the interfacial contact of inorganic perovskite and metal oxide transport layers
		$CsPbI_3$-A'X' or $FAPbI_3$-A'X' intermediate phases promoting the formation of their pure black perovskite phases

intramolecular exchange reaction of solvent and AX, or full penetration of AX into PbX_2 film with increased grain boundaries and pore density (Table 1).

Two-step methods for PSCs achieved the first certified record PCE in 2013,[38] proving their advantages in fabricating high-performance PSCs. Two-step methods in solution process refer to a sequential deposition of a first-step PbX_2 layer by spin-coating and a second-step AX layer by dipping, spin-coating, or evaporation (Figure 2A).[39] Incomplete transformation of PbX_2 into perovskite phase often occurs (Figure 2B).

DMSO has a higher D_N than DMF, resulting from a shorter Pb–O bond length of 2.386 Å (2.431 Å for DMF) in coordination with PbI_2.[40,41] The strong coordinative DMSO as a solvent with PbI_2 results in a complete transformation into $MAPbI_3$ perovskite film with a narrow distribution of crystal sizes and low surface roughness.[42] In 2015, Yang et al. found that DMSO and PbI_2 in precursor solution can react to form two different intermediate phases, $PbI_2(DMSO)_2$ and $PbI_2(DMSO)$. This uniform and dense pre-deposited PbI_2-DMSO film can directly convert into $FAPbI_3$ as the intercalated DMSO can be easily replaced by FAI (Figure 2C). In addition, the $FAPbI_3$ film derived from PbI_2-DMSO exhibited a dense and well-developed grain structure with larger grains than the film derived from PbI_2. The device based on this intramolecular exchange method achieved a certified PCE of 20.1%.[43] A closely packed $PbI_2(DMSO)_x$ ($0 \leq x \leq 1.86$) film was further prepared by adding DMSO into PbI_2-DMF solution, and an intramolecular exchange of MAI with DMSO enabled the complexes to form an ultra-flat and dense $MAPbI_3$ thin films.[27] Bing et al. drop-casted MAI-FAI-isopropanol solution on top of a PbI_2-CsI-DMSO-DMF precursor film.[25] This DMSO-complex intermediate phase slowed down perovskite crystallization and was fully converted to a uniform and dense $Cs_{0.15}(MA_{0.7}FA_{0.3})_{0.85}PbI_3$ film with highly ordered orientation.

Cheng et al. demonstrated that *N*-Methylpyrrolidone (NMP) is a superior coordination additive, forming only one intermediate compound in a wide operation window of a Pb^{2+}/NMP ratio between 1 and 4.[44] Jo et al. also obtained PbI_2-NMP

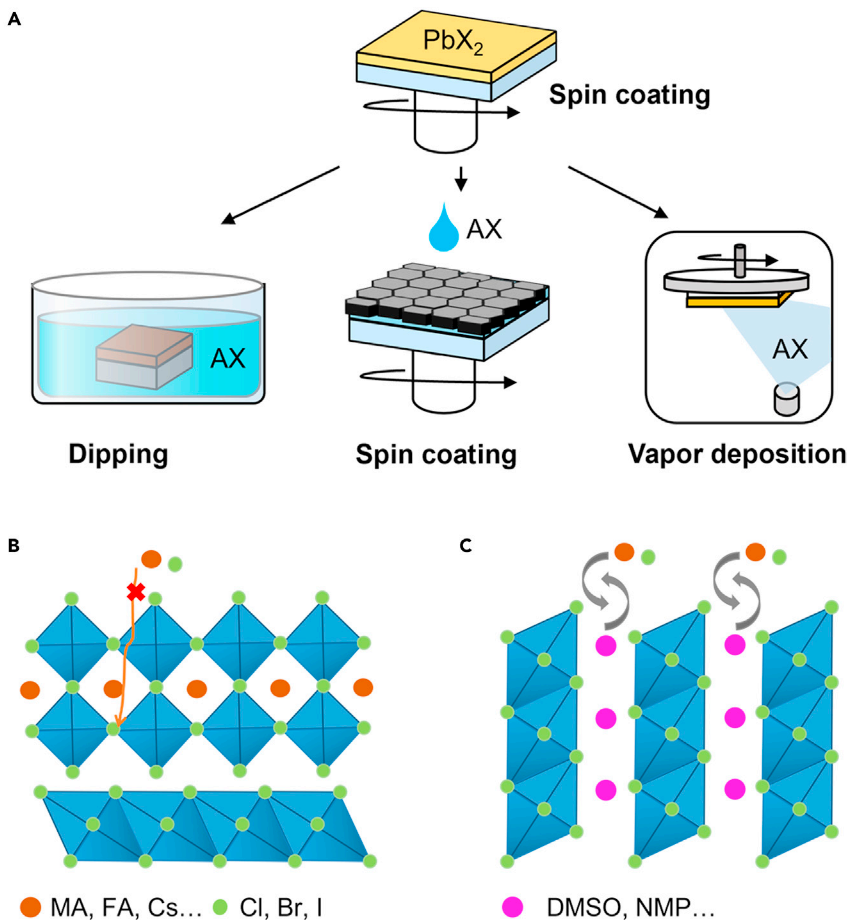

Figure 2. IPE in two-step methods for perovskite film formation

(A) Scheme of two-step methods for perovskite films.

(B) Scheme of incomplete transformation of PbX_2 into perovskite phase in the conventional two-step method.

(C) Intramolecular exchange reaction of PbX_2-solvent complex and A cations.

intermediate phase by mixing PbI_2 with NMP solvent.[45] The PbI_2-NMP facilitated the formation of nearly perfect surface coverage over the substrate, which might be due to the enhanced solubility of PbI_2/NMP in DMF solution than PbI_2 alone. The grain size of FAI-based perovskite film derived from PbI_2-NMP is up to ≈ 1 μm, which is obviously larger than those derived from PbI_2 or PbI_2-DMSO. Films with the largest average grain size from PbI_2-NMP intermediate contributed to the best PCE of studied solar cells. Dimethylacetamide (DMAC), possessing a stronger electron-pair-donating ability with PbI_2 compared with DMF, has also been introduced into PbI_2/DMF solution to form the PbI_2-DMAC intermediate. The PbI_2-DMAC film subsequently converts into a uniform and dense $MAPbI_3$ perovskite film by molecular exchange between MAI and DMAC.[46] Zhang et al. compared pyridine (Py), 4-*tert*-butylpyridine (*t*BP), DMF, DMSO, ethylenediamine (EDA) as ligands (L) to form perovskite films via a judicious design of PbI_2:$(L)_x$ intermediates. Calculation shows that the activation energies of the studied intermediate phases to perovskites are lower than that of PbI_2 to perovskite.[47] PbI_2:$(Py)_2$ film shows a uniform nanoporous morphology, which ensured its complete conversion to $MAPbI_3$ by molecule exchange between MAI and ligands. Compared with Py, *t*BP, DMSO, DMF, and EDA, acetonitrile (ACN) shows weaker coordination ability (lower D_N) and easier evaporation nature. Therefore, ACN as an additive into PbI_2/DMF solution leads to the reduction of ACN-PbI_2-DMF cluster size and an as-formed film with increased grain boundaries and pore density, which improves the nucleation rate in MAI penetration. ACN-based precursor also facilitates the perovskite grain growth for the quick removal of solvents. With simultaneous control of the relative rates for nucleation and grain growth, the corresponding perovskite film shows gradually enlarged grain size in the range of a few hundred nanometers to over 1 μm by increasing ACN concentration.[48] Hui et al. reported the synthesis of stable black-phase α-$FAPbI_3$ with an ionic liquid solvent methylamine formate (MAFa).[26] MAFa has a strong interaction with PbI_2 through C=O···Pb chelation and N–H···I hydrogen bonds, which promotes the vertical growth of PbI_2 with respect to the substrate. Nanoscale channels in the films lower the permeation barrier of FAI and enable the transformation to α-$FAPbI_3$ over a broad range of relative humidity (RH, from 20% to 90%) and temperature (25°C–100°C). The PSCs with PCEs over 24% were achieved. The unencapsulated cells display high stability, retaining 80% and 90% of their initial efficiencies for 500 h at 85°C and continuous light stress, respectively.

IPE based on perovskite precursors and solvents

Intermediate phases in perovskite precursors (ABX_3) and solvents differ from those in PbX_2 and solvents due to the stronger interaction of AX toward PbX_2.[49] Intermediate phases of perovskite precursors and solvents have been widely studied in one-step solution method, MA gas post-healing or solvation process, and antisolvent method.

One-step solution method

In one-step solution method, intermediate phase (ABX_3-solvent) can adjust the crystallization to form uniform perovskite films or increase the crystallinity of perovskite films (Table 1).

It has been reported that a mixture of PbI_2 and MAI in hot DMF solution can produce a dinuclear plumbate intermediate $(MA)_2(PbI_3)_2 \cdot DMF_2$, which easily loses DMF to produce $MAPbI_3$ under nitrogen at 20°C–100°C.[50] The PbI_2-MAI reaction in DMSO is similar to that in DMF reaction, except that a mixture of PbI_2 and MAI in hot DMSO produces a new class of trinuclear plumbate as fibers $(MA)_2[(PbI_3)_2PbI_2] \cdot DMSO_2$.[50] Petrov et al. identified three different intermediate phases in PbI_2-MAI in DMF solution, PbI_2-rich $(MA)_2(DMF)_2Pb_3I_8$, stoichiometric $(MA)_2(DMF)_2Pb_2I_6$, and MAI-rich $(MA)_3(DMF)PbI_5$ (Figure 3A). DMSO intermediate $(MA)_2(DMSO)_2Pb_3I_8$ (Figure 3B) is more stable with a formation enthalpy of −390.4 kcal/mol than that of DMF-intermediate $(MA)_2(DMF)_2$ Pb_3I_8 with −180.1 kcal/mol.[51] It again explains the preferential precipitation of DMSO-adduct rather than DMF adduct when a mixture of DMF and DMSO is used as a solvent. Likewise, Chen et al. found that a small amount of DMSO incorporation into DMF (volume ratio of 0.93:0.07) renders PSCs with enhanced PCE performance, which they attributed to the formation of the more stable intermediate phase MAI-PbI_2-DMSO in DMF/DMSO system compared with DMF-based devices.[52]

Bu et al. reported a PbI_2-templated crystallization strategy to fabricate antisolvent-free and ambient-air-printed high-performance $(FACs)PbI_3$-based solar cells by introducing NMP.[53] They found that in $(FACs)PbI_3$ perovskite DMF solution, solvent-coordinated perovskite intermediate phases of $Cs_2Pb_3I_8 \cdot 4DMF$ and $FA_2Pb_3I_8 \cdot 4DMF$ can be easily converted to δ-$(FACs)PbI_3$ and results in poor film morphology. Upon the addition of NMP, the as-formed PbI_2-NMP restrained the formation of $Cs_2Pb_3I_8$-4DMF and $FA_2Pb_3I_8$-4DMF intermediate phases, leading to a successful conversion into dense α-$(FACs)PbI_3$ films even at RT (Figure 3C). It is because when PbI_2-NMP reacts with

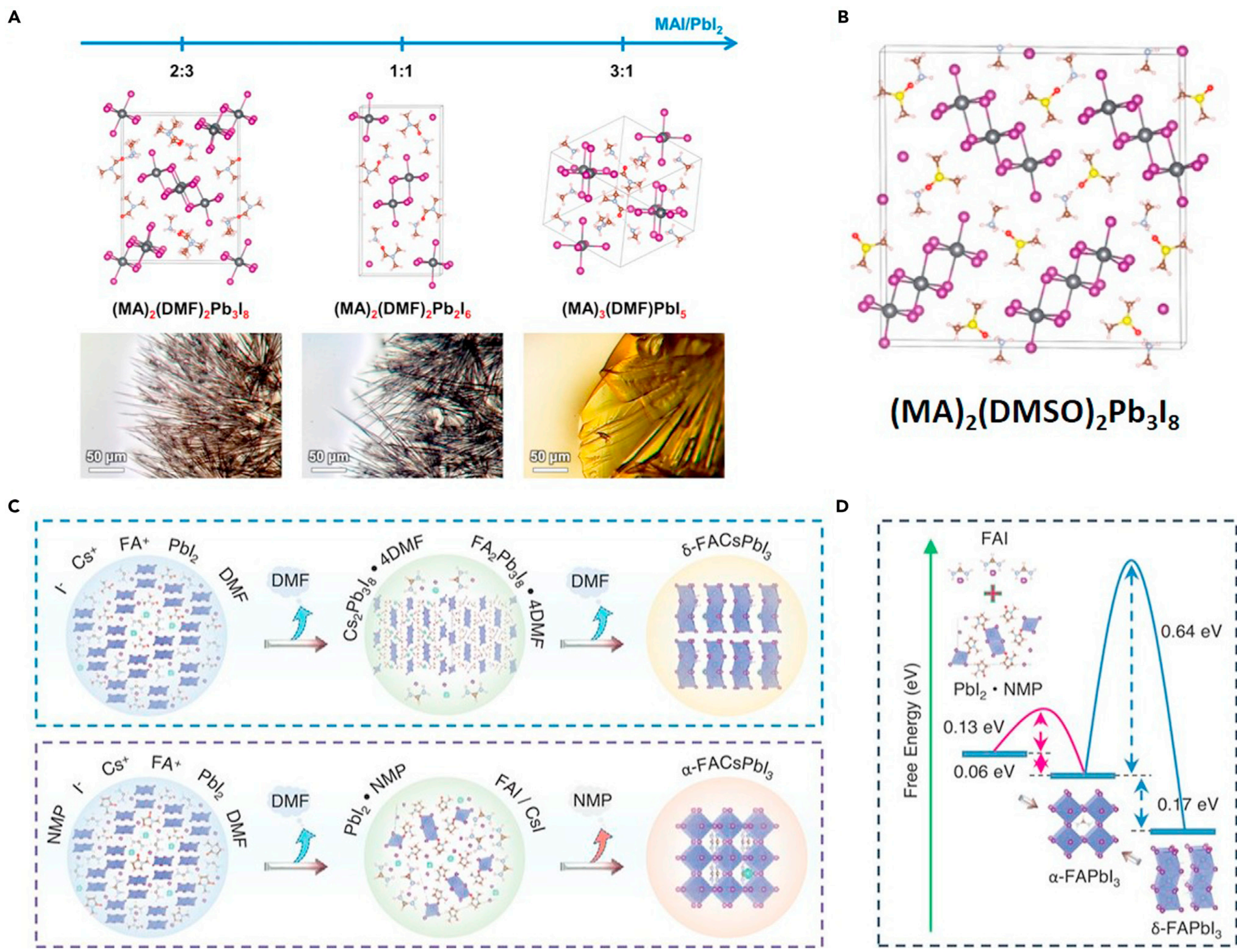

Figure 3. IPE in one-step solution methods

(A) Three different intermediate phases (PbI_2-rich $(MA)_2(DMF)_2Pb_3I_8$, stoichiometric $(MA)_2(DMF)_2Pb_2I_6$, and MAI-rich $(MA)_3(DMF)PbI_5$) in PbI_2-MAI precursor in DMF solution.

(B) Crystal structures of $(MA)_2(DMSO)_2Pb_3I_8$ intermediate phase. Copyright from ACS Publications (2017).[51]

(C) Schematic diagram of crystal growth with or without NMP in one-step solution method.

(D) Free-energy calculation for the formation of $FAPbI_3$ perovskites with or without NMP in one-step solution method. Copyright from AAAS (2021).[53]

FAI to form perovskite, the formation energy of α-$FAPbI_3$ is significantly lower than the direct δ-to-α $FAPbI_3$ phase conversion (Figure 3D). By further introducing excess $PbCl_2$, the formation of PbX_2-0.5NMP-0.5DMF adduct further suppresses the growth of $FA_2Pb_3I_8$-4DMF nuclei and results in a denser perovskite film.

With conventional spin-coating methods following post-annealing, inorganic perovskite films often exhibit inferior film morphology featuring incomplete coverage on substrates or undesirable interfacial contact, which may be originated from the rapid evaporation of DMSO residual and subsequent fast nucleation and growth of perovskite crystals at high annealing temperature.[54] Adjusting the intermediate phase can effectively alter the nucleation rate of perovskite crystals. Intermediates PbI_2-DMSO and $PbBr_2$-DMSO were used as precursors to slow down the crystallization for high-quality $CsPbI_2Br$ films with large crystalline grains and long carrier lifetime.[55] Such photoactive $CsPbI_2Br$ perovskites can even be formed at RT via these intermediate phases.[56] By changing the DMF/DMSO ratio in the precursor solution, the obtained

CsI-PbX_2-DMSO intermediate phase during spin-coating can not only significantly lower the annealing temperature to 100°C but also regulate the crystal growth kinetics for high-quality $CsPbI_2Br$ films.[57,58] In addition, residual solvent trapped in precursor films was reported to control the intermediate phase evolution during crystallization process. By resting a freshly spin-coated $CsPbI_{3-x}Br_x$ film for tens of minutes in a N_2 glovebox, a majority of DMSO was removed, which avoided the irregular growth and severe aggregation of perovskite crystals during high-temperature annealing. As a result, the annealed perovskite films are pinhole free with crystal size above 5 μm.[54]

2D perovskites contain bulky organic cations, which provide a steric barrier for surface water adsorption and thus enhance the stability of perovskite materials.[59] The choice of solvents is also important for high-quality 2D perovskite films. Because of the rapid evaporation of DMF and subsequent rapid homogeneous nucleation, pure DMF solvent results in a random crystal orientation of perovskites. However, when DMF is mixed with DMSO, a crystal orientation perpendicular to the substrate can be achieved via intermediate phases.[60,61] Tuning the solvent composition with a mixture DMF and DMSO ratio of 1:1 leads to the growth of highly oriented 2D perovskite films with much improved optoelectronic properties than that prepared with pure DMF, corresponding to >80% efficiency improvement.[61]

IPE is also applied in large-scale manufacturing (such as doctor blade and slot-die coating methods) of perovskite devices for commercial applications. By introducing volatile solvents of 2-methoxyethanol (2-ME) and ACN into coordinative DMSO, DMF solvents, uniform perovskite films with grain size up to 2 μm by blade coating at an unprecedented speed of 99 mm/s was realized at RT, producing perovskite modules with efficiency above 16% with an aperture area of over 60 cm^2.[62] It was later found that a limited amount of DMSO prevents the formation of a crystalline intermediate phase related to $MAPbI_3$ and 2-ME and induces the formation of the $MAPbI_3$ perovskite phase. Higher DMSO content leads to the precipitation of the $(DMSO)_2MA_2Pb_3I_8$ intermediate phase that negatively affects the thin-film morphology.[63]

Besides DMSO, γ-butyrolactone (GBL) is also used as a solvent component to modify the perovskite film. By mixing DMF and GBL as solvents, smooth perovskite films with uniform crystal domains can be obtained, which also improves the interfacial contacts on poly(3,4-ethylenedioxythiophene) polystyrene sulfonate (PEDOT:PSS) substrate.[64]

Ionic liquid MAAc is used as a special solvent for dense, pinhole-free perovskite films without antisolvent. Its high viscosity protects the perovskite from moisture invasion from RH = 30% to 80%. MAAc was further used for all-inorganic $CsPbI_{3-x}Br_x$ perovskite films. The dissolution of PbX_2 is through the formation of strong N–H⋯I hydrogen bonds and strong Pb–O bonds with perovskite precursors. The strong interactions stabilize the inorganic perovskite precursor solution and allow production of inorganic perovskite films with low trap density by retarding the crystallization. The surface roughness of the MAAc film was 10.5 nm, much lower than that of the DMF/DMSO film (14.1 nm). It leads to the high-quality devices with PCE of 17.10% and V_{oc} over 1.30 V.[65] For 2D perovskites, a mixture of DMSO and MAAc was adopted to dually manage the crystallization kinetics of Ruddle-Popper Sn perovskite films by Lewis adducts from DMSO and the ion exchange from MAAc. The formation of intermediates BAMASn–Ac (BA: butylamine) facilitates the ion exchange between I^- and Ac^-, which then yields dense $BA_2MA_3Sn_4I_{13}$ films. This homogeneous 2D film exhibited an average grain size of 9 μm and low electron and hole defect densities with a magnitude of 10^{16}.[66]

MA gas post-healing or solvation process

Due to the low boiling point (6°C) and strong coordination ability with Pb^{2+}, MA has been employed as a post-healing gas to heal raw perovskite films or as a special "solvent" to solvate the perovskite precursors for high-quality perovskite films. The liquid intermediate phase $MAPbI_3$-xMA can promote the formation of uniform perovskite films as the quick evaporation of MA gas leads to the formation of over-saturation and then large amount of crystal nucleus (Table 1).

In 2015, Zhou et al. first reported the MA-induced defect-healing (MIDH) method to deposit high-quality $MAPbI_3$ perovskite films.[67] MA gas with basic N atoms reacts with the PbI_6-octahedra framework in raw $MAPbI_3$ perovskite, resulting in the complete collapse of perovskite structure into a non-crystalline and photo-inactive $MAPbI_3$-xMA liquid intermediate phase (Figure 4A). It is further found that MA-MA^+ dimers are formed via hydrogen bonds in $MAPbI_3$-xMA intermediates with 1D-PbI_3 (δ-phase).[68] The MA-MA^+ dimers with a larger ionic radius (445 pm) compared with MA^+ cations (217 pm) as A-site cation resulted in a complete collapse of original α-phase $MAPbI_3$ with 3D-PbI_3^- units and the formation of thermodynamically unstable 1D-PbI_3^-. The weak bonding between MA and $MAPbI_3$ could dissolve $MAPbI_3$ completely and result in spontaneous and rapid transition of δ-α phase under ambient condition due to the easy release of MA gas (Figure 4B).[68] In the case of mesoscopic-oxide layer on the substrate, the liquid is likely to evenly spread and infiltrate readily into the mesoporous structure and lead to perfect interfacial contact. MA molecules are then released from the liquid upon the reduction of MA gas partial pressure, which results in a volume contraction and the reconstruction of the $MAPbI_3$ perovskite structure into an ultra-smooth and dense film (Figure 4C). The incorporation and escaping of MA molecules are completely reversible.

The importance of MA gas in the fabrication of $MAPbI_3$ thin films was further highlighted by comparing different amines. The lower basicity and the smaller size of NH_3 compared with MA does not cause the complete collapse of the solid into a liquid intermediate phase at RT. However, the larger-molecular amine gases, such as ethylamine or n-butylamine, would lead to the irreversible formation of a stable non-$MAPbI_3$ phase. The 1D NH_4PbI_3 precursor film can react with MA gas to initially form raw $MAPbI_3$ perovskite film due to molecular exchange. Such raw $MAPbI_3$ film then transforms into a $MAPbI_3$-xMA liquid intermediate phase by further absorbing excess MA and finally into a uniform $MAPbI_3$ film by degassing MA gas.[69] When adopting $HPbI_3$ as a precursor, it is observed that H^+ determines the stoichiometric formation of $MAPbI_3$ perovskite by reacting with MA gas and reconstructing the PbI_6 octahedra. $MAPbI_3$-3.5MA as the liquid intermediate phase was detected, which results in ultra-smooth and full-coverage $MAPbI_3$ perovskite thin films.[70] Such MIDH method is virtually independent of the morphology of the starting precursor films, making this deposition process highly robust, and is especially benign for large-scale perovskite films prepared by blade coating such as slot-die.[71,72] Furthermore, it has been verified that MA cannot transform inorganic perovskite $CsPbX_3$ into a liquid intermediate phase as the MA molecules can hardly break the ···Cs–X–Pb··· chemical bonds.[36] As a result, a certain amount of MAX (X = I or Br) additives are introduced into $CsPbX_3$ precursor, which accelerates the absorption of MA gas and leads to the formation of a liquid intermediate phase and uniform $CsPbX_3$ films after thermal annealing.[36] Fan et al. reported the MA-assisted growth of (110) uniaxial-oriented perovskite by controlling the pressure of MA gas in a closed chamber,

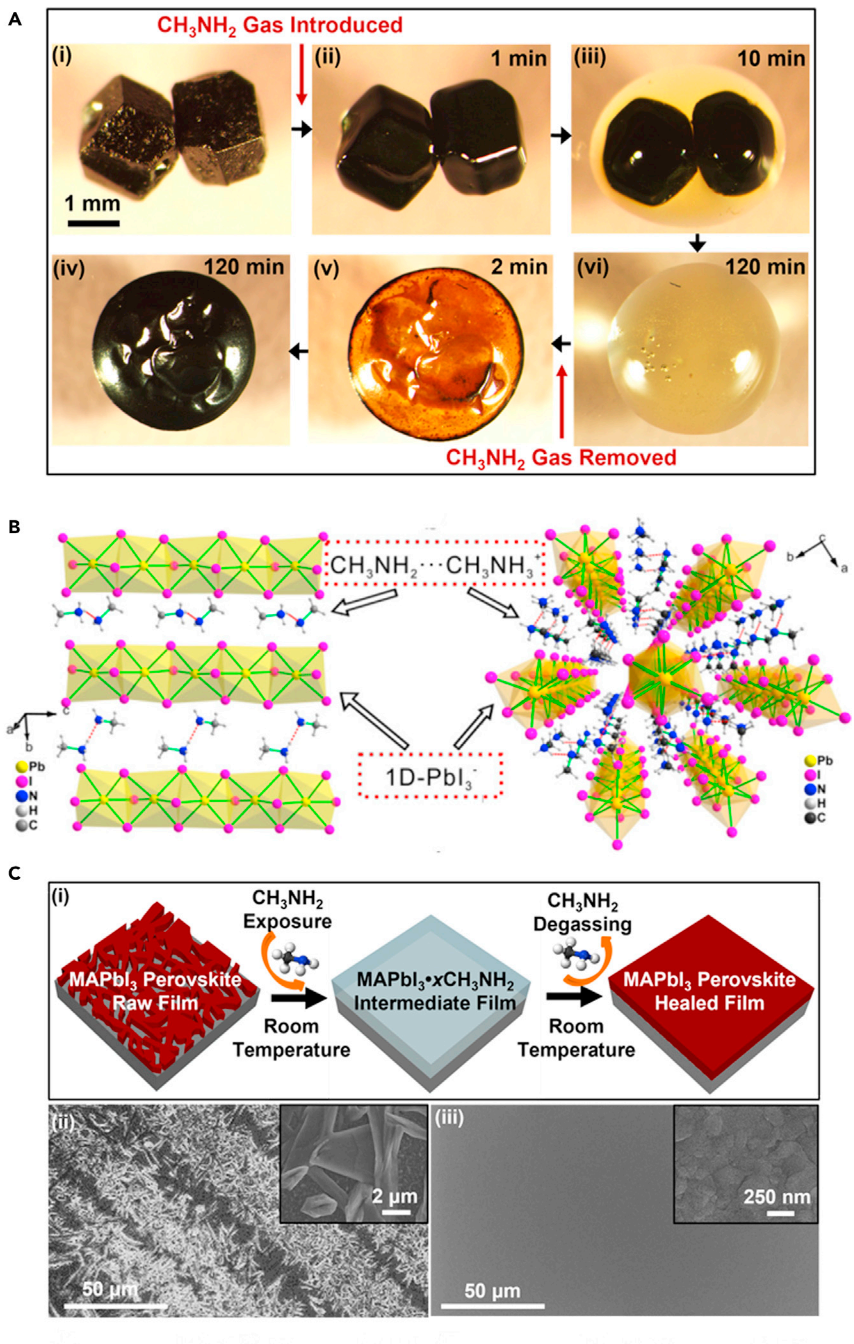

Figure 4. Methylamine gas post-healing toward perovskite films

(A) *In situ* optical microscopy of the morphology evolution of two $MAPbI_3$ perovskite crystals upon exposure to MA gas and MA degassing. (i) Before MA gas exposure, (ii) MA gas introduced, (iii) partial collapse of perovskite structure and conversion to liquid, (iv) full conversion to liquid, (v) MA degassing, and (vi) perovskite back-conversion completed. Copyright from Wiley Online Library (2015).[67]

(B) Structure of a MA-$MAPbI_3$ single crystal from the MA-gas-pre solution. Copyright ACS Publications (2020).[68]

(C) (i) Scheme of MA-induced defect-healing method of $MAPbI_3$ films. Top-view SEM images of $MAPbI_3$ films: (ii) raw film and (iii) healed film, (insets: higher magnification images). Copyright from Wiley Online Library (2015).[67]

where MA gas can be *in situ* generated within a temperature range between 100°C and 180°C and intaken into the raw $MAPbI_3$ film to form liquid intermediate phase of $MAPbI_3$-$x$$MA^0$.[73] The slow release of MA gas from liquid phase induces low nucleation sites and takes more time for crystal growth, resulting in homogeneous nucleation and millimeter-sized grains. As a result, the corresponding PSCs achieved a PCE of 21.4%, which is among the highest reported values for $MAPbI_3$-based devices.

Chen et al. reported a deposition route for $MAPbI_3$ perovskite films that used neither solvent or vacuum.[72] Specifically, they prepared amine complex precursors MAI-3MA and PbI_2-MA by the direct reaction between MAI or PbI_2 powder and MA gas. These amine complex precursors were then mixed stoichiometrically to fabricate perovskite films by a pressure processing method which helps to spread the liquid precursor under the polyimide substrate film. The key reaction processes are displayed below:

$$\mathrm{MAI} + 3\,\mathrm{MA} = \mathrm{MAI} - 3\mathrm{MA} \tag{Equation 1}$$

$$\mathrm{PbI_2} + \mathrm{MA} = \mathrm{PbI_2} - \mathrm{MA} \tag{Equation 2}$$

$$\mathrm{MAI} - 3\mathrm{MA} + \mathrm{PbI_2} - \mathrm{MA} \rightarrow \mathrm{MAPbI_3} + 4\,\mathrm{MA} \tag{Equation 3}$$

The deposited perovskite films were free of pinholes and highly uniform and exhibited grain size distribution about 0.8–1.0 μm which is 3–4 times larger than that of the conventional spin-coating processed films. A module based on mesoporous architecture fabricated in air at low temperature reached a certified PCE of 12.1% with an active area of 36 cm^2.

ACN/MA solvent by bubbling MA gas into a polar and aprotic solvent ACN is employed to dissolve the $MAPbI_3$ precursor which was unable to be dissolved in pure ACN.[74] The dissolution of the perovskite precursors is carried out by the MA gas itself through the formation of a liquid perovskite phase ($MAPbI_3$-xMA). The ACN primarily acts as a conduit of MA and this intermediate compound, which is miscible with the ACN. This method combined the merits of both MA treatments and the solution processing, which allows the obtainment of extremely smooth and pinhole-free $MAPbI_3$ films with crystal grain sizes ranging between 500 and 700 nm. Furthermore, this method guarantees the uniformity of perovskite films on substrates with an area of 125 cm^2, which is extremely promising for the upscaling of high-efficiency, solution-processed devices.

Wang et al. reported a nonionic precursor by dissolving the $MAPbI_3$ perovskite crystals into a monomethylamine (MMA) tetrahydrofuran solution (here MMA is the same as MA discussed above), followed by dilution with ACN.[75] MMA gas can insert between $[PbI_6]^{4-}$ octahedrals by hydrogen bonding and forming the intermediate phase of $MA(MMA)_nPbI_3$. Different from the free mobile ions in the ionic precursors (e.g., MAI and PbI_2), the intermediate states capture these ions and demonstrate the nonionic properties, which can realize immediate perovskite crystallization at RT after spin-coating. The obtained film displays superior (110) preferred orientation and an extraordinarily large carrier diffusion length of 4.6 μm. As a result, the corresponding PSCs exhibit an efficiency of 21.8% based on $MAPbI_3$ with superior moisture stability.

Antisolvent method

In 2014, Jeon et al. used toluene drop-casting antisolvent technique to obtain extremely uniform and dense perovskite layers of $MAPb(I_{1-x}Br_x)_3$ (x = 0.1–0.15)

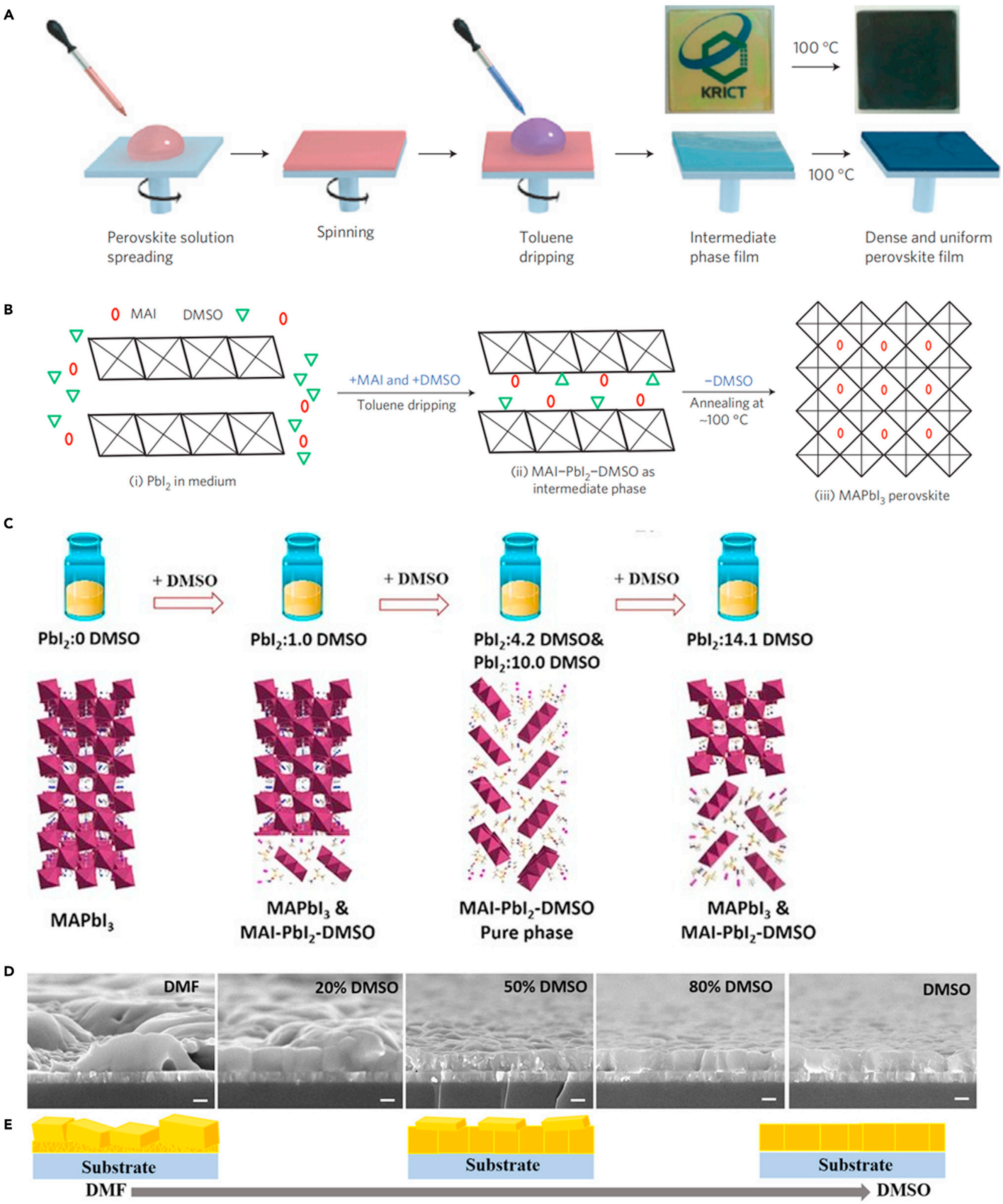

Figure 5. Antisolvent method and involved intermediate phases

(A) Antisolvent method for depositing perovskite film.

(B) Scheme for the formation of perovskite phase via IPE. (1) PbI_2 layers. (2) MAI and DMSO guest molecules intercalated between the layers, forming the flat MAI-PbI_2-DMSO intermediate phase film when toluene is introduced. (3) The intermediate phase film converting to the perovskite phase by extracting DMSO molecules via thermal-annealing process. Copyright from Nature Publishing Group (2014).[21]

Figure 5. *Continued*

(C) Schematic illustrating the relation between the amount of DMSO and the component of intermediate film. Copyright from Elsevier (2017).[76]
(D and E) Cross-sectional SEM images (scale bar, 200 nm) and schematic representation of film formation of the $FAPbBr_3$ films with different DMSO concentrations. Copyright from Wiley Online Library (2021).[77]

from a mixed solvent of GBL and DMSO via a MAI-PbI_2-DMSO intermediate phase (Figures 5A and 5B).[21] Since then, antisolvent method has been widely employed in one-step solution process method. The type and amount of the coordinated solvents used for perovskite precursor solution, and the type of antisolvents, are all showing fundamental impacts on the formation of the intermediate phases and final perovskite films. These intermediate phases can prolong the processing window of antisolvent dropping for uniform perovskite film, and the formation of pure intermediate phase can further increase the quality of perovskite films (Table 1).

Amount of coordinated solvents

It has been reported that there is a narrow processing window of dropping antisolvent when using pure-DMF-based perovskite precursor solutions.[78] The usage of DMF/DMSO solvent mixture enables a wider window of dropping antisolvent for film processing, as DMSO has a higher D_N than DMF and can coordinate with Pb^{2+} more strongly.[22] Solvent composition influences the formation of intermediate phases and therefore the growth of perovskite crystal. For example, equimolar DMSO (MAI/PbI_2/DMSO=1:1:1 mol %) forms MAI-PbI_2-DMSO intermediate phase and high-quality perovskite films using diethyl ether (DE) as antisolvent. Precursors with excess DMSO such as $PbI_2 \cdot MAI \cdot 2DMSO$ (1:1:2 mol %) result in extra PbI_2 in the annealed film. This may result from the partial formation of $PbI_2(DMSO)_2$ intermediate in the fresh film from excess DMSO precursor.[79]

Bai et al. also found that a pure and stable intermediate phase is key to growing aligned and vertically monolithic perovskite crystals.[76] By varying the concentration of DMSO, the intermediate film changed from pure perovskite to sequentially perovskite/$MA_2Pb_3I_8(DMSO)_2$ mixture, pure $MA_2Pb_3I_8(DMSO)_2$ and $MA_2Pb_3I_8(DMSO)_2$/perovskite mixture (Figure 5C). They found pure $MA_2Pb_3I_8(DMSO)_2$ intermediate phase tended to grow along the (110) direction to form aligned perovskite crystals with few horizontal grain boundaries and reduced trap state density, which allowed to fabricate NiO-based inverted PSCs with a PCE of 18.4%, along with little hysteresis and high stability.

During the fabrication of $FAPbBr_3$ film, Hu et al., for the first time, identified a FABr-$PbBr_2$-DMSO intermediate phase and synthesized its single crystal.[77] The investigation of phase evolution in the film formation process reveals that DMSO favors the growth of perovskite crystals in a Volmer-Weber mode,[80] featuring a closely packed island. Thus, higher fraction of DMSO in the precursor solutions enables the crystallization of the FABr-$PbBr_2$-DMSO intermediate phase and modulates the crystallization process for a smooth $FAPbBr_3$ perovskite film (Figures 5D and 5E).

Types of coordinated solvents

NMP solvent with a large D_N and high boiling prolongs the processing window for antisolvent dropping. The precursor film from $MAPbI_3$/NMP solution after spin-coating can also be immediately immersed in a bath of antisolvent, e.g., DE, to extract the precursor-NMP, which then induces rapid crystallization of uniform and ultra-smooth perovskite thin films at RT.[81]

Lee et al. replaced DMSO with NMP to obtain a stable intermediate FAI-PbI_2-NMP phase, which can be converted into a uniform and pinhole-free $FAPbI_3$ film.[82] The

C=N stretch vibration peak of bare FAI is slightly shifted in the FAI-PbI_2-NMP intermediate, but not for the FAI-PbI_2-DMSO, indicating NMP interacts with FAI. Moreover, the interaction between FA cation and solvents through hydrogen bond exhibits the interaction energy of -1.253 and -1.407 eV for DMSO and NMP, respectively. Thus, they concluded a stronger interaction between NMP with the FA cation than DMSO, which facilitates the formation of the stable intermediate phase and subsequently the flat, uniform, and pinhole-free films. Based on the molecular interactions, they proposed several criteria for selecting the Lewis bases: (1) high hydrogen-bond-accepting ability and low hydrogen bond donor ability, such as NMP, (2) sterically accessible electron donating atom, the oxygen in NMP is more sterically accessible than that of DMSO, and (3) matching of the hardness and softness of the corresponding Lewis acid and base. As NMP is a softer base compared with DMSO, FA cation, a softer Lewis acid than MA cation, preferentially interacts with NMP over DMSO, whereas MA cation preferentially interacts with DMSO over NMP.[82]

Dimethylpropyleneurea (DMPU) as a high-D_N (44 kcal/mol) solvent additive compared with DMSO in DMF perovskite precursor solution can prolong the processing window of antisolvent dropping to form uniform perovskite film with large grains.[22] Tetramethylene sulfoxide (TMSO) as an additive was also found to form a strong coordination bond with perovskite precursors, which results in a stable intermediate-state complex film, long processing window for antisolvent dropping and suppressed $Cs_{0.1}FA_{0.85}MA_{0.05}PbI_{2.85}Br_{0.15}$ perovskite nuclei.[83] In addition, TMSO can restrain spontaneous α-δ phase transition and extend the annealing window (the storage time of the intermediate-state films during annealing) to as long as 20 min. Thus, TMSO in the precursor film promotes the formation of high-quality perovskite films with increased grain size and surface texture. Feng et al. proposed that dimethyl sulfide (DS) as a solvent additive is more capable of providing electrons to coordinate with Pb^{2+} than DMSO, as the sulfur atoms in DS have smaller electronegativity than oxygen atoms in DMSO.[84] In addition, the relatively stronger polarity and the smaller Gibbs free energy of DS molecule than DMSO with PbI_2 is more beneficial to form stable intermediates. These features dramatically slow down the rate of perovskite crystallization and result in large grains ($\approx$409 nm compared with $\approx$151 nm without DS) with better crystallinity. The retarded transformation kinetics during the film crystallization process leads to reduced trap density in the $MAPbI_3$ perovskite film by an order of magnitude compared with the one without DS.

Types of antisolvent

Formation of a pure intermediate phase for the fresh film after antisolvent treatment is beneficial for high-quality perovskite films and high-efficiency PSCs, which is relevant to the polarity of antisolvent.[24,85] It has been verified that two key factors influence the quality of perovskite film: the solubility of the organic precursors in the antisolvent and the miscibility of the host solvent and antisolvent.[23] Chlorobenzene (CB), anisole (ANS), DE, diisopropyl ether (DIE), dibutyl ether (DBE) (the trend of polarity: ANS > CB > DE > DIE > DBE) have been compared as antisolvents. Among them, precursor perovskite film treated with DIE antisolvent can yield a purer intermediate phase (perovskite-DMSO), which subsequently induces the crystallization of large grains and a smooth surface.[24] CB and ANS are miscible with both DMF and DMSO due to their slightly higher polarities. They might remove excessive DMF and DMSO and lead to the formation of perovskite phase in the fresh film. Upon thermal annealing, the intermediate phases with low purities are more likely to have smaller grains and form more inhomogeneous films. The intermediate phase treated with DE exhibits an obvious X-ray diffraction (XRD) signal for $PbI_2(DMSO)_2$ as DE can extract DMF more quickly from the precursor solution and leave more redundant DMSO to coordinate with PbI_2 forming $PbI_2(DMSO)_2$. It changes the stoichiometric ratio of the intermediate phase and may impede the crystallization of the pure perovskite phase. The polarity of DBE is too low to be miscible with DMF, resulting in DMF not being completely removed through antisolvent treatment. During annealing process, the rapid evaporation of residual DMF from the intermediate phase can disturb the phase formation and morphology of the final perovskite film. By utilizing DIE as a green antisolvent, both double- and triple-cation FA-based PSCs achieved excellent performance, with 21.3% efficiency in triple-cation PSCs.[24]

For inorganic perovskites, the traditional antisolvents, such as chlorobenzene, toluene, and DE, are difficult to fully remove excess DMSO/DMF in fresh film for a uniform perovskite film. It results from the strong interaction of Cs^+ and DMSO/DMF.[86] However, the $CsPbI_3$ perovskite film derived from methyl acetate (MeOAc) or ethyl acetate (EtOAc) antisolvent exhibits compact, pinhole-free films with large grain domains (up to 10 μm). This phenomenon cannot be well explained by the typically discussed solvent parameters (boiling point, vapor pressure, miscibility, and polarity) and the degree of removal of DMSO. Since Cs^+ as a soft acid will preferentially bind to soft bases, the antisolvent interactions with Cs^+ are expected to follow the trend of softness (MeOAc ≈ EtOAc > ANS > DE) based on the polarizability of the Lewis basic oxygens. It promotes the heterogeneous nucleation process and MeOAc or EtOAc antisolvents facilitate the formation of high-quality $CsPbI_3$ perovskite films derived from Cs^+-antisolvents (MeOAc or EtOAc) intermediate phases.

IPE based on perovskite precursors and volatile additives

Volatile additives of organic ammonium salts or acids A′X′ (where A′ is H^+, NH_4^+, MA^+, FA^+, DMA^+, etc.; X′ is Cl^-, Br^-, I^-, Ac^-, NO_3^-, HPO_3^-, SCN^-, etc.[2,29–32,87,88]) often have strong coordination ability with perovskite precursors and form intermediate phases ABX_3-A′X′ without coordinated solvent molecules in freshly prepared films. Generally, these volatile additives no longer exist in the final perovskite films after thermal annealing.

The ways of introducing volatile additives A′X′ in perovskite precursor solutions do not make any difference, if the overall compositions of final precursor solutions are identical.[33] For example, the perovskite precursor solutions with $NH_4Ac:MAI:PbI_2$ = 1:1:1, $MAAc:NH_4I:PbI_2$ = 1:1:1 and $MAI:NH_4I:PbAc_2:PbI_2$ = 1:1:0.5:0.5 have the same ion components despite the different ion sources, which leads to the similar properties for the target films.[33] Volatile additives play important roles mainly in three aspects: morphology and trap density control of perovskite films, improving interfacial contact, and formation of photoactive perovskite phases (Table 1).

Morphology and trap density control of perovskite films

Amount of additives

The amount of volatile additives in precursor solutions is crucial in controlling the morphology and trap density of perovskite films. Lee et al. added 2 mol MACl into 1 mol $MAPbI_3$ precursor to prepare the precursor solution with $PbCl_2$:MAI = 3:1.[28] The halide perovskite $MAPbI_{3-x}Cl_x$ can fully fill in the meso-superstructured substrate and the corresponding PSCs achieved a PCE of 10.9%. To study the effects of additive amount, different amounts of MACl are introduced into $MAPbI_3$ precursor solutions by two routes as presented in the following equations[89,90]:

$$(1 + x)MAI + (1 - x/2)PbI_2 + (x/2)PbCl_2 = MAPbI_3 + xMACl \text{ or}$$

$$MAI + PbI_2 + xMACl = MAPbI_3 + xMACl$$

Both show that $x = 1$ achieved the best coverage of $MAPbI_3$ perovskite-capping layer. Besides, the MACl:$MAPbI_2Br$ = 1:1 precursor solution yields the uniform $MAPbI_2Br$ film and better coverage on substrate than other ratios.[91] In this study, MACl as a soft template controls the initial formation of a solid solution with main $MAPbI_2Br$ precursor components.[91] Cl^- ions facilitate the release of excess MA^+ at a relatively low annealing temperature, which is critical to slow down the perovskite crystallization process and thus improve the growth of the crystal domains during annealing.[92] The elemental analysis reveals negligible amount of Cl atoms in the $MAPbI_{3-x}Cl_x$ film. Therefore, the formation of the $MAPbI_{3-x}Cl_x$ perovskite is likely driven by the release of gaseous MACl from an intermediate organometal mixed halide phase.

For inorganic perovskites, we added 0.5 mol FAAc into 1 mol $CsPb(I_{0.75}Br_{0.25})_3$ precursor solution to obtain more uniform perovskite films than other FAAc content (0.25, 0.75, and 1 mol). It shows that the optimized additive content in inorganic perovskite is different from that in OIHP.[31]

Types of additives

MAAc as an additive was first reported in 2015 for preparing ultra-smooth and pinhole-free perovskite films from $PbAc_2$:MAI = 3:1 precursor solution.[93] As the decomposition temperature of MAAc is lower than those of MACl and MAI, MAAc is thermally unstable and much easier to be removed during annealing process. Thus, under the same annealing temperature, the nucleation density is much higher for the films processed with the MAAc additive. It results in the formation of a large amount of relatively small crystals in a short timescale (1–2 min) and full coverage over the substrate. In contrast, for approaches using MACl or MAI as additive, longer annealing time is required to remove these additives, which leads to low nucleation density, large crystals, and discontinuous perovskite films.[93] Even though the uniformity of films can be guaranteed by MAAc, the grain size of perovskite is much smaller due to weaker crystallographic orientation, as is expected from the fast rate of crystallization. Blended additive engineering is therefore deployed to regulate perovskite films, such as MAAc and MACl co-additives, where MAAc ensures film smoothness and coverage, and MACl facilitates perovskite film crystallinity.[94] Moore et al. studied the kinetics of perovskite formation from the intermediate phases based on four additives (MAI, MACl, MAAc, and $MANO_3$).[16] The change in activation energy (E_a) of the intermediate-to-perovskite transition is dominated by the removal of these additives from the intermediate. Specifically, the chloride system has a lower E_a (86.6 kJ/mol) than iodide system and can be proceed at lower temperatures. It provides additional time for grain growth and expands the coarsening window right before decomposition. As a result, the addition of MACl leads to the enhanced transport properties and charge carrier lifetimes of $MAPbI_3$ films and improved device performance.

Ke et al. showed that adding a small amount of $Pb(SCN)_2$ in the perovskite precursor can significantly increase the grain size and improve the crystallinity of $MAPbI_3$ thin films without significant incorporation of SCN^- ions into the resultant films.[95] The evaporation of MASCN during annealing leads to the formation of PbI_2 at the grain boundaries, which reduces the shunt current along these grain boundaries. As a result, PSCs with 5 mol % $Pb(SCN)_2$ additive significantly reduced hysteresis and increased fill factor.

NH_4Cl was used as an additive to form more uniform perovskite films than MACl additive on the PEDOT:PSS substrate.[96] Si et al. depicted the process of NH_4I-assisted controlling of $MAPbI_3$ perovskite nucleation and crystal growth.[97] Smaller NH_4^+ (compared with MA^+) diffuses into the $[PbI_6]^{4-}$ octahedral layer to form NH_4PbI_3 intermediate phase and results in extra heterogeneous nucleation sites. Because of the stronger bonding with $[PbI_6]^{4-}$ than NH_4^+, MA^+ replaces NH_4^+ by ions exchange, resulting in the formation of high-quality $MAPbI_3$ perovskite with enlarged grain size and lowered defect density. For 2D PSCs, Yang et al. rerouted crystallization pathway to suppress defects in vertically oriented 2D perovskite by using NH_4Cl and NH_4I mixed additives.[98] The specific adsorption of the ammonium halide on different perovskite crystal planes allows the restrain of the dynamic preferred growth of (111) planes, while the minority (202) planes emerge as secondary nucleation sites to stimulate the creation of large grains. As a result, a vertically oriented 2D perovskite film with high homogeneity, reduced trap density of states, and good carrier kinetics was formed.

Halide acids as volatile additives have also been studied to improve the quality of perovskite films.[99] For example, the PbI_2 + MAI + HCl precursor solution yielded a smooth $MAPbI_3$ film simply by spin-coating.[100] HBr aqueous solution additive enhanced the solubility of $MAPbBr_3$ in DMF and retarded the nucleation during spin-coating. As a result, $MAPbBr_3$ in DMF + HBr solution yielded a $MAPbBr_3$ perovskite layer with higher surface coverage.[101] Likewise, HI additive increases the solubility of $MAPbI_3$ in DMF thereby facilitating the formation of pinhole-free $MAPbI_3$ thin film without any impurities. The improved perovskite film morphology increases the charge diffusion length and lifetime in devices.[102]

Interfacial contact

Due to the weak interfacial interaction with metal oxide charge transport layer, inorganic halide perovskites have a big interfacial contact issue compared with OIHP.[31] Pinholes are observed at the interface of inorganic perovskite and TiO_2 or SnO_2 substrates. To solve this problem, different volatile additives have been introduced into inorganic perovskite precursor solutions. The annealed $CsPb(I_{0.75}Br_{0.25})_3$ films with 0.5FAAc, 0.5MAAc, and $0.5FAI_{0.24}Br_{0.76}$ additives show good interfacial contacts, while pinholes still exist with $0.5NH_4Ac$ and 0.5BAAc additives. It is interpreted that FA^+ or MA^+ can be incorporated into perovskite lattice to form 3D OIHP intermediate phase, which plays important roles for improving the interfacial contact (Figures 6A–6C).[31]

Formation of photoactive perovskite phases

At RT, the desirable $FAPbI_3$ or $CsPbI_3$ black perovskite phases can spontaneously transform into their more thermodynamically stable "yellow" orthorhombic δ-phases. The black $FAPbI_3$ and $CsPbI_3$ phases can be obtained at high temperatures, e.g., over 310°C for $CsPbI_3$ and 160°C for $FAPbI_3$. In fact, these black perovskite films can be stable at RT in a dry ambient, resulting from the substrate-clamping-driven texture which creates large biaxial strain to stabilize the black phase.[105] However, the high energy barrier from yellow phase to black phase for these films restrains the δ-to-α phase transformation. Formation of intermediate phase by introducing additives can reduce such formation energy during the phase transformation process and promote the formation of pure black phase.[29,35,53]

In early studies, MACl and FACl have been applied in the synthesis of α-$FAPbI_3$. The 1:1 ratio of $FAPbI_3$:additive in the precursor solution leads to the formation of intermediate phase of $FAPbI_3$-MACl or $FAPbI_3$-FACl, which avoids the formation of

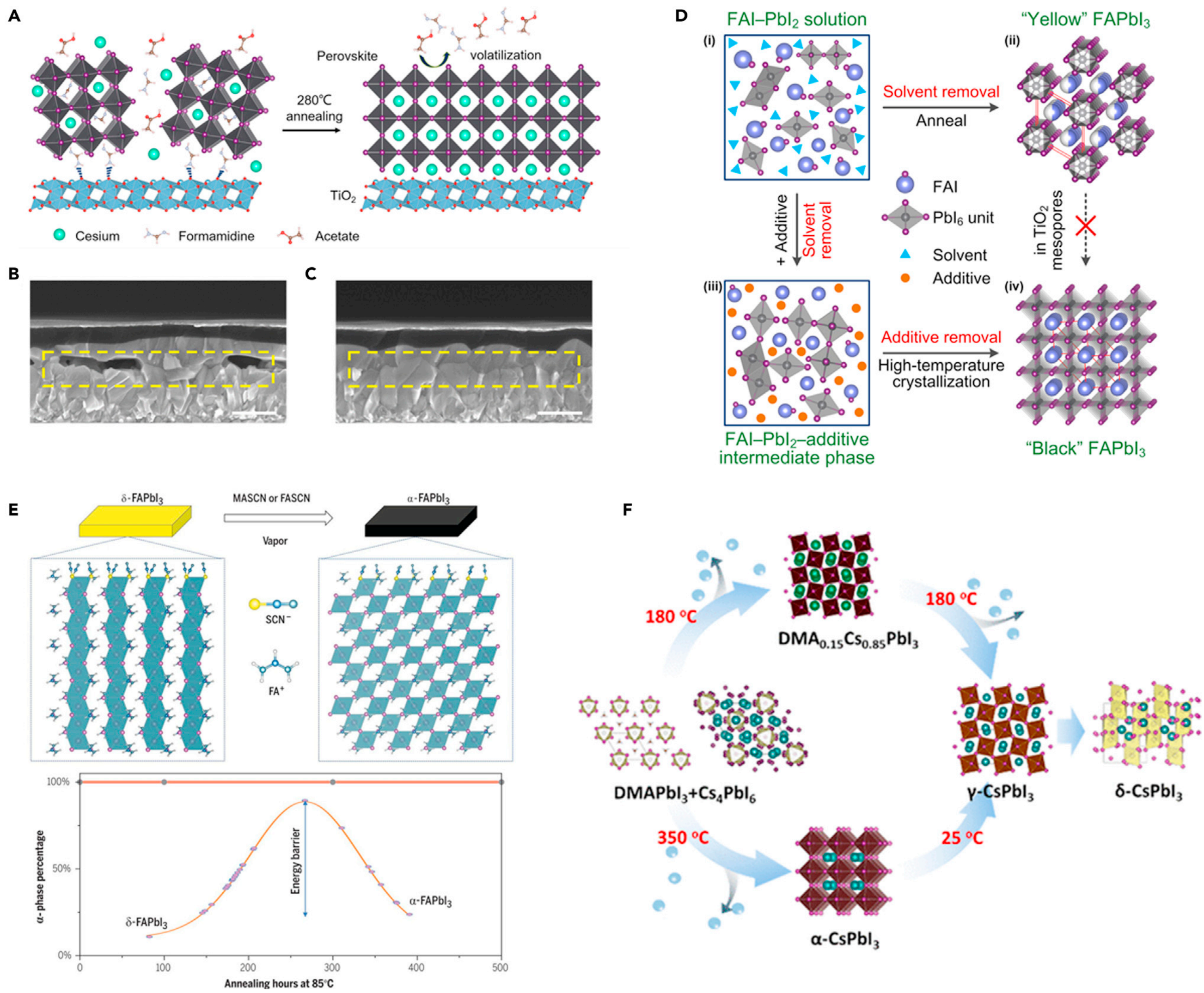

Figure 6. Formation of photoactive perovskite phases via IPE

(A) Schematic illustration of the IPE process of a $CsPb(I_{0.75}Br_{0.25})_3$ film with FAOAc additive.

(B and C) Cross-sectional SEM images of $CsPb(I_{0.75}Br_{0.25})_3$-based planar structure solar cells without (B) and with (C) FAOAc additive, respectively. Scale bar, 500 nm. The yellow dashed boxes show the perovskite/metal oxide interfaces. Copyright from Elsevier publications (2020).[30]

(D) Schematic illustration of the mechanisms for the evolution of $FAPbI_3$ phases on mesoporous TiO_2 scaffolds. Copyright from ACS Publications (2015).[35]

(E) Stable and phase pure MASCN vapor-treated $FAPbI_3$ films annealed at 85°C for 500 h in an N_2 environment. Copyright from AAAS (2021).[103]

(F) Chemical composition and phase evolution in DMAI-derived $CsPbI_3$. Copyright from ACS Publications (2019).[104]

yellow phases and directly transforms into the 3D black $FAPbI_3$ perovskite phase along with the removal of MACl or FACl during annealing process (Figure 6D).[35]

Kim et al.[29] found that the formation energy of α-$FAPbI_3$ structure is related to the amount of MACl in the precursor solution, and the MACl-assisted phase transformation of pure α-$FAPbI_3$ can even occur without annealing process. They revealed the roles of MACl additive on the perovskite crystal structure in three aspects: (1) the p orbital localization of I^- induced by Cl^- can strengthen the interactions of FA^+ and I and improve the stability of the $FAPbI_3$ perovskite structure; (2) the incorporation of MA^+ contributes to the total volume shrinkage in the MA^+-incorporated system by ≈ 0.57%, compared with the bare α-phase $FAPbI_3$. In addition, the dipole effect of the MA^+ contributes to the lower formation energy of the MA system due to the ten times more dipole moment of MA^+ relative to that of FA^+; (3) the optimal concentration of MACl depends on the doping formation energy of MACl into the α-$FAPbI_3$ structure in the pre-annealing step and the formation energy of the perovskite structure including MAI in the post-annealing step. Notably, the formation energy is stabilized between 33% and 44%, corresponding to the best MACl concentration of 40%.

Lu et al. reported a deposition method using MASCN or FASCN vapor treatment to convert δ-$FAPbI_3$ to the desired pure α-phase below the thermodynamic phase-transition temperature (100°C).[103] In their study, yellow δ-$FAPbI_3$ film was first formed by conventional spin-coating method. The as-formed film was annealed at 100°C for 1 min and subjected to a MASCN vapor environment for ~5 s until the color changed to black. Due to the strong affinity to Pb^{2+} ions, SCN^- anions can displace the iodides to form intermediate phase with Pb^{2+} on the surface of δ-$FAPbI_3$. It induces the rearrangement of ions at the surface of δ-$FAPbI_3$ and facilitates the phase transition to the edge-sharing intermediates and further corner-sharing α-$FAPbI_3$. The formation of α-$FAPbI_3$ layer on the top surface templates the δ-to-α phase transition toward the bulk. Once the pure α-$FAPbI_3$ is finally formed, its reverse conversion to the δ-phase is inhibited by a high energy barrier (Figure 6E).

IPE method is also showing significant advantages for preparing pure inorganic $CsPbI_3$ black perovskite phases. Initially, the stable and efficient $CsPbI_3$ solar cells were fabricated by adding a small amount of HI into a $CsPbI_3$-DMF precursor solution or using $HPbI_3$ as an alternative precursor to PbI_2.[102,106–109] However, Ke et al. unveiled that the exact composition of $CsPbI_3$ annealed at a temperature lower than 180°C is actually $Cs_{1-x}DMA_xPbI_3$, where DMA^+ is derived from hydrolyzing DMF with a small amount of water in the solution.[110] It is further suggested that DMA^+ is doped into the perovskites in the form of DMAI/$DMAPbI_x$ intermediate compound, whose amount is proportional to the HI content.[111,112] Since then, DMA^+ as a popular additive has been widely used for preparing $CsPbI_3$ inorganic perovskite thin films. By varying the CsI/$DMAPbI_3$ ratio in precursor solutions, high-quality $Cs_{1-x}DMA_xPbI_3$ ($0 \leq x \leq 1$) perovskite films can be prepared.[113] The CsI/DMAI molar ratio was found to determine the $CsPbI_3$ perovskite phase annealed at 210°C, e.g., $CsPbI_3$-0.5DMAI and $CsPbI_3$-0.75DMAI precursors form γ-$CsPbI_3$, while β-$CsPbI_3$ was obtained from $CsPbI_3$-DMAI and $CsPbI_3$-1.5DMAI.[32] The β-$CsPbI_3$ film annealed at 210°C with an E_g of 1.68 eV was treated with choline iodide and then constructed PSCs which exhibit a reproducible PCE of 18.4%.[30] Meng et al. verified that $CsPbI_3$-DMAI can transform into thermodynamically stable $DMA_{0.15}Cs_{0.85}PbI_3$ (E_g = 1.67 eV) film via a solid-solid reaction between $DMAPbI_3$ and Cs_4PbI_6 at an annealing temperature of 180°C. By extending the annealing time, γ-$CsPbI_3$ and δ-$CsPbI_3$ film can be obtained sequentially (Figure 6F).[104] They further reported the fabrication of $CsPbI_3$ perovskite films with high crystallinity by adding DMAI and excess CsI in the perovskite precursor solution using leaching method at RT. This precursor can form 1D $DMAPbI_3$ and 0D Cs_4PbI_6 in the raw films. The recrystallization of perovskite phase occurs simultaneously with the dissolution of CsI and DMAI compounds from the solid raw film during leaching process, and then photoactive $CsPbI_3$ phase was formed at RT. DMAI can suppress the formation of δ-$CsPbI_3$ in raw film, while excess amount of CsI can compensate the inevitable loss of CsI during the leaching treatment.[114] They also proposed a Cs_4PbI_6-mediated method for synthesizing $Cs_{0.85}FA_{0.15}PbI_3$ perovskites.[115] It is found that FAI-0.85CsI-PbI_2 precursor first forms an FA-rich perovskite phase $FA_xCs_{1-x}PbI_3$ and Cs_4PbI_6 owing to their lower free energy at low temperature. Cs_4PbI_6 intermediate phase has a low Cs cation diffusion barrier and therefore offers a fast ion exchange with the preformed FA-rich perovskite phase to finally form the Cs-rich $FA_xCs_{1-x}PbI_3$ perovskite. The formation of the Cs_4PbI_6 in advance can efficiently restrict the formation of the undesired δ-$CsPbI_3$ in the perovskite films.[115]

Moreover, MAI is also introduced into $CsPbI_3$ precursor solutions as an additive. After annealing at 330°C for 20 min in N_2 glovebox, $CsPbI_3$-MAI first forms a $MAPbI_3$-based perovskite template, then transforms into pure γ-$CsPbI_3$ film by further ion exchange of MA^+ ions in the $MAPbI_3$ with Cs^+.[116] $CsPbI_3$ perovskite film could also be obtained by annealing the precursor film (stoichiometric CsAc + MAI + PbI_2) under low crystallization temperature of 100°C.[117]

CONCLUSION AND OUTLOOK

As has been summarized in this review, the IPE method is very facile and effective for depositing high-quality halide perovskite films with high phase purity. The crystallization dynamics of perovskite films can be readily tuned by modifying the components of intermediate phases in the fresh film. Therefore, the IPE method has been widely applied in fabricating highly efficient PSCs with various film-processing techniques. Despite that the tremendous progress has been made, the understanding of roles and mechanisms for IPE is still in its infancy. The accurate crystal structures and compositions of PbX_2-solvent intermediate phases in two-step method have been rarely reported other than PbX_2-DMSO. The perovskite films prepared from PbX_2-solvent intermediate phases for high-efficiency PSCs often possess excess PbI_2 residue,[118] even on the surface of perovskite films, which cannot be explained by the intramolecular exchange reaction. In addition, the composition and distribution of the formed perovskite films from two-step method with intermediate phases are far from clearance, as these films often show noticeably different band gaps with regard to those prepared using one-step method with the same chemical formula.[118–120]

Likewise, only the structures and crystal growth processes of ABX_3-DMSO and ABX_3-DMF intermediate phases have been studied in one-step solution method.[51] Some solvents indeed improve the quality of perovskite films by forming new intermediate phases, whose composition, structures, and phase transformation process needs thorough exploration. Moreover, FA-based perovskites with higher thermal stability and more appropriate band gaps than MA-based perovskite for single-junction photovoltaics do not compatible with MIDH method as post-healing an FA-based perovskite film by MA gas leads to the irreversible formation of a yellow phase, which is remarkably different from $MAPbI_3$-xMA system.[121] The underlying reasons should be investigated in order to apply MIDH method in fabricating FA-based perovskite films with high quality.

As for ABX_3-A′X′ intermediate phases, the choice of A′X′ additive is still empirical with insufficient clear mechanisms as guidance. For example, it is reported that the best film coverage can be achieved from both OIHP precursors and inorganic perovskite precursors, although the molar ratios of additives in these two types of precursors are quite different.[31] The associated mechanisms deserve further exploration combining with their crystallization kinetics.

To elucidate the roles and mechanisms of IPE for high-quality perovskite films, the following techniques are recommended. XRD is essential for unraveling the crystal structures of intermediate phases. *In situ* characterizations (such as *in situ* XRD, *in situ* transient absorption) are necessary to study the crystallization kinetic during the formation and transition of intermediate phases. Meanwhile, theoretical supports are highly needed to the related fundamental understandings in the future.

For the fabrication of stable, efficient, and reproducible PSCs, the following aspects should also be considered via IPE.

(1) The development of new coordinating solvents that are green and controllable during IPE film processing. Despite the vastly used solvents such as DMF, DMSO, and NMP, the performances of obtained perovskite films and solar cells are still highly dependent on environmental conditions and are sometimes significantly different among different labs. The choice of solvents should be based on their D_N to coordinate with precursors, which is important to form stable intermediate phases. In addition, other properties such as volatility, viscosity, polarity, and vapor pressure of solvents should also be taken into consideration when using IPE method.

(2) Volatile additives with appropriate hydrogen bond interaction or Lewis acid-base reaction with perovskite precursors provide another huge library of choices. The competitions among organic component, solvent, and additive to coordinate with PbX_2, and the intermediate phase reaction should be reversible to facilitate the continuous component exchange. As such, the relationship between the solvent/additive (D_N, amount, etc.) and the precursors in forming stable and optimum intermediate phases for converting into desirable perovskite films needs to be further explicit, which guarantees the reproducibility of high-performance perovskite devices. Blended additive (or multi-additives) can be further considered to fully utilize their own advantages.[122]

(3) In general, intermediate phase includes a dynamic balance process, which changes with the standing time, annealing temperature, atmosphere, etc. Accurate adjustment of the intermediate phase process can promote the formation of high-quality perovskite films. For example, in MIDH method, the slow release of MA gas from liquid intermediate phase can adjust the process of intermediate phase and induce more time for crystal growth, resulting in homogeneous nucleation and millimeter-sized grains.[73] With further understanding of the mechanism and crystallization kinetics of intermediate phase process, accurate manipulation of intermediate phase process will be expected to achieve further improvement of perovskite film quality and efficiencies of devices.

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AUTHOR CONTRIBUTIONS

W.X. and Z.W. initiated this review, conducted the literature review, and drafted the manuscript. Z.W. and W.X. framed the review. J.Z., Z.W., and W.X. contributed to all figures and graphical abstract and revised the manuscript. S.(F).L., A.H., and S.A. contributed to the revision of this manuscript.

DECLARATION OF INTERESTS

The authors declare no competing interests.

REFERENCES

1. Xu, J.X., Boyd, C.C., Yu, Z.J., Palmstrom, A.F., Witter, D.J., Larson, B.W., France, R.M., Werner, J., Harvey, S.P., Wolf, E.J., et al. (2020). Triple-halide wide-band gap perovskites with suppressed phase segregation for efficient tandems. Science *367*, 1097–1104.
2. Min, H., Kim, M., Lee, S.U., Kim, H., Kim, G., Choi, K., Lee, J.H., and Seok, S.I. (2019). Efficient, stable solar cells by using inherent bandgap of alpha-phase formamidinium lead iodide. Science *366*, 749–753.
3. Xiang, W., Liu, S., and Tress, W. (2021). A review on the stability of inorganic metal halide perovskites: challenges and opportunities for stable solar cells. Energy Environ. Sci. *14*, 2090–2113.
4. Yi, C., Liu, C., Wen, K., Liu, X.K., Zhang, H., Yu, Y., Fan, N., Ji, F., Kuang, C., Ma, B., et al. (2020). Intermediate-phase-assisted low-temperature formation of γ-CsPbI3 films for high-efficiency deep-red light-emitting devices. Nat. Commun. *11*, 4736.
5. Liu, Y., Cui, J., Du, K., Tian, H., He, Z., Zhou, Q., Yang, Z., Deng, Y., Chen, D., Zuo, X., et al. (2019). Efficient blue light-emitting diodes based on quantum-confined bromide perovskite nanostructures. Nat. Photonics *13*, 760–764.
6. Zhuang, R., Wang, X., Ma, W., Wu, Y., Chen, X., Tang, L., Zhu, H., Liu, J., Wu, L., Zhou, W., et al. (2019). Highly sensitive X-ray detector made of layered perovskite-like (NH4)3Bi2I9 single crystal with anisotropic response. Nat. Photonics *13*, 602–608.
7. Pan, W., Wu, H., Luo, J., Deng, Z., Ge, C., Chen, C., Jiang, X., Yin, W.-J., Niu, G., Zhu, L., et al. (2017). Cs2AgBiBr6 single-crystal X-ray detectors with a low detection limit. Nat. Photon. *11*, 726–732.
8. NREL. Best Research-Cell Efficiency Chart; 2021.
9. Kojima, A., Teshima, K., Shirai, Y., and Miyasaka, T. (2009). Organometal halide perovskites as visible-light sensitizers for photovoltaic cells. J. Am. Chem. Soc. *131*, 6050–6051.
10. Green, M.A., Dunlop, E.D., Hohl-Ebinger, J., Yoshita, M., Kopidakis, N., and Hao, X. (2021). Solar cell efficiency tables (version 58). Prog. Photovolt. *29*, 657–667.
11. NREL (2020). Champion photovoltaic module efficiency chart. https://www.nrel.gov/pv/module-efficiency.html.
12. Grancini, G., Roldán-Carmona, C., Zimmermann, I., Mosconi, E., Lee, X., Martineau, D., Narbey, S., Oswald, F., De Angelis, F., Graetzel, M., and Nazeeruddin, M.K. (2017). One-year stable perovskite solar cells by 2D/3D interface engineering. Nat. Commun. *8*, 15684.
13. Al-Ashouri, A., Köhnen, E., Li, B., Magomedov, A., Hempel, H., Caprioglio, P., Márquez, J.A., Morales Vilches, A.B., Kasparavicius, E., Smith, J.A., et al. (2020). Monolithic perovskite/silicon tandem solar cell with >29% efficiency by enhanced hole extraction. Science *370*, 1300–1309.
14. Jošt, M., Al-Ashouri, A., Lipovšek, B., Bertram, T., Schlatmann, R., Kaufmann, C.A., Topič, M., and Albrecht, S. (2020). Perovskite/CIGS tandem solar cells - can they catch up with perovskite/c-Si tandems? 47th IEEE Photovoltaic Specialists Conference (PVSC), 0763–0766.
15. Brenner, T.M., Egger, D.A., Kronik, L., Hodes, G., and Cahen, D. (2016). Hybrid organic–inorganic perovskites: low-cost semiconductors with intriguing charge-transport properties. Nat. Rev. Mater. *1*, 15007.
16. Moore, D.T., Sai, H., Tan, K.W., Smilgies, D.M., Zhang, W., Snaith, H.J., Wiesner, U., and Estroff, L.A. (2015). Crystallization kinetics of organic-inorganic trihalide perovskites and the role of the lead anion in crystal growth. J. Am. Chem. Soc. *137*, 2350–2358.
17. Jung, M., Ji, S.G., Kim, G., and Seok, S.I. (2019). Perovskite precursor solution chemistry: from fundamentals to photovoltaic applications. Chem. Soc. Rev. *48*, 2011–2038.
18. Dunlap-Shohl, W.A., Zhou, Y., Padture, N.P., and Mitzi, D.B. (2019). Synthetic approaches for halide perovskite thin films. Chem. Rev. *119*, 3193–3295.
19. Lee, J.-W., Kim, H.-S., and Park, N.-G. (2016). Lewis acid–base adduct approach for high efficiency perovskite solar cells. Acc. Chem. Res. *49*, 311–319.
20. Li, Y., Zhi, L., Ge, G., Zhao, Z., Cao, X., Chen, F., Cui, X., Lin, F., Ci, L., Sun, J., et al. (2018). Investigation on crystallization of CH3NH3PbI3 perovskite and its intermediate phase from polar aprotic solvents. Cryst. Growth Des. *19*, 959–965.
21. Jeon, N.J., Noh, J.H., Kim, Y.C., Yang, W.S., Ryu, S., and Seok, S.I. (2014). Solvent engineering for high-performance inorganic–organic hybrid perovskite solar cells. Nat. Mater. *13*, 897–903.
22. Hamill, J.C., Schwartz, J., and Loo, Y.-L. (2018). Influence of solvent coordination on hybrid organic–inorganic perovskite formation. ACS Energy Lett *3*, 92–97.
23. Taylor, A.D., Sun, Q., Goetz, K.P., An, Q., Schramm, T., Hofstetter, Y., Litterst, M., Paulus, F., and Vaynzof, Y. (2021). A general approach to high-efficiency perovskite solar cells by any antisolvent. Nat. Commun. *12*, 1878.
24. Wang, L.Y., Wang, X., Deng, L.L., Leng, S.B., Guo, X.J., Tan, C.H., Choy, W.C.H., and Chen, C.C. (2020). The mechanism of universal green antisolvents for intermediate phase controlled high-efficiency formamidinium-based perovskite solar cells. Mater. Horiz. *7*, 934–942.
25. Bing, J.M., Kim, J., Zhang, M., Zheng, J.H., Lee, D.S., Cho, Y.Y., Deng, X.F., Lau, C.F.J., Li, Y., Green, M.A., et al. (2019). The impact of a dynamic two-step solution process on film formation of Cs0.15(MA0.7FA0.3)0.85PbI3 perovskite and solar cell performance. Small *15*, e1804858.
26. Hui, W., Chao, L., Lu, H., Xia, F., Wei, Q., Su, Z., Niu, T., Tao, L., Du, B., Li, D., et al. (2021). Stabilizing black-phase formamidinium perovskite formation at room temperature and high humidity. Science *371*, 1359–1364.
27. Li, W., Fan, J., Li, J., Mai, Y., and Wang, L. (2015). Controllable grain morphology of perovskite absorber film by molecular self-assembly toward efficient solar cell exceeding 17. J. Am. Chem. Soc. *137*, 10399–10405.
28. Lee, M.M., Teuscher, J., Miyasaka, T., Murakami, T.N., and Snaith, H.J. (2012). Efficient hybrid solar cells based on meso-superstructured organometal halide perovskites. Science *338*, 643–647.
29. Kim, M., Kim, G.-H., Lee, T.K., Choi, I.W., Choi, H.W., Jo, Y., Yoon, Y.J., Kim, J.W., Lee, J., Huh, D., et al. (2019). Methylammonium chloride induces intermediate phase stabilization for efficient perovskite solar cells. Joule *3*, 2179–2192.
30. Wang, Y., Dar, M.I., Ono, L.K., Zhang, T., Kan, M., Li, Y., Zhang, L., Wang, X., Yang, Y., Gao, X., et al. (2019). Thermodynamically stabilized β-CsPbI3-based perovskite solar cells with efficiencies >18. Science *365*, 591–595.
31. Zhang, J., Wang, Z., Mishra, A., Yu, M., Shasti, M., Tress, W., Kubicki, D.J., Avalos, C.E., Lu, H., Liu, Y., et al. (2020). Intermediate phase enhances inorganic perovskite and metal oxide interface for efficient photovoltaics. Joule *4*, 222–234.
32. Wang, Y., Liu, X., Zhang, T., Wang, X., Kan, M., Shi, J., and Zhao, Y. (2019). The role of dimethylammonium iodide in CsPbI3 perovskite fabrication: additive or dopant? Angew. Chem. Int. Ed. Engl. *58*, 16691–16696.
33. Li, C.W., Wang, Z.W., Chang, Y., Zong, Y.X., Ji, F.X., Zhang, B., Li, H.Y., and Pang, S.P. (2016). Insight into the effect of ion source for the solution processing of perovskite films. RSC Adv *6*, 85026–85029.
34. Yoo, J.J., Seo, G., Chua, M.R., Park, T.G., Lu, Y., Rotermund, F., Kim, Y.K., Moon, C.S., Jeon, N.J., Correa-Baena, J.P., et al. (2021). Efficient perovskite solar cells via improved carrier management. Nature *590*, 587–593.
35. Wang, Z., Zhou, Y., Pang, S., Xiao, Z., Zhang, J., Chai, W., Xu, H., Liu, Z., Padture, N.P., and Cui, G. (2015). Additive-modulated evolution of HC(NH2)2PbI3 black polymorph for mesoscopic perovskite solar cells. Chem. Mater. *27*, 7149–7155.
36. Shao, Z., Wang, Z., Li, Z., Fan, Y., Meng, H., Liu, R., Wang, Y., Hagfeldt, A., Cui, G., and Pang, S. (2019). A scalable methylamine gas healing strategy for high-efficiency inorganic perovskite solar cells. Angew. Chem. Int. Ed. Engl. *58*, 5587–5591.
37. Gutmann, V. (1976). Solvent effects on the reactivities of organometallic compounds. Coord. Chem. Rev. *18*, 225–255.
38. Burschka, J., Pellet, N., Moon, S.-J., Humphry-Baker, R., Gao, P., Nazeeruddin, M.K., and Grätzel, M. (2013). Sequential deposition as a

route to high-performance perovskite-sensitized solar cells. Nature *499*, 316–319.

39. Wang, Z., Zhang, J., Guo, W., Xiang, W., and Hagfeldt, A. (2021). Formation and stabilization of inorganic halide perovskites for photovoltaics. Matter *4*, 528–551.

40. Wakamiya, A., Endo, M., Sasamori, T., Tokitoh, N., Ogomi, Y., Hayase, S., and Murata, Y. (2014). Reproducible fabrication of efficient perovskite-based solar cells: X-ray crystallographic studies on the formation of CH3NH3PbI3 layers. Chem. Lett. *43*, 711–713.

41. Miyamae, H., Numahata, Y., and Nagata, M. (1980). The crystal structure of lead(II) iodide-dimethylsulphoxide(1/2), PbI2(DMSO)2. Chem. Lett. *9*, 663–664.

42. Wu, Y., Islam, A., Yang, X., Qin, C., Liu, J., Zhang, K., Peng, W., and Han, L. (2014). Retarding the crystallization of PbI2 for highly reproducible planar-structured perovskite solar cells via sequential deposition. Energy Environ. Sci. *7*, 2934–2938.

43. Yang, W.S., Noh, J.H., Jeon, N.J., Kim, Y.C., Ryu, S., Seo, J., and Seok, S.I. (2015). SOLAR CELLS. High-performance photovoltaic perovskite layers fabricated through intramolecular exchange. Science *348*, 1234–1237.

44. Cheng, F., Jing, X., Chen, R., Cao, J., Yan, J., Wu, Y., Huang, X., Wu, B., and Zheng, N. (2019). N-Methyl-2-pyrrolidone as an excellent coordinative additive with a wide operating range for fabricating high-quality perovskite films. Inorg. Chem. Front. *6*, 2458–2463.

45. Jo, Y., Oh, K.S., Kim, M., Kim, K.-H., Lee, H., Lee, C.-W., and Kim, D.S. (2016). High performance of planar perovskite solar cells produced from PbI2(DMSO) and PbI2(NMP) complexes by intramolecular exchange. Adv. Mater. Inter. *3*, 1500768.

46. Cao, X., Zhi, L., Li, Y., Fang, F., Cui, X., Yao, Y., Ci, L., Ding, K., and Wei, J. (2017). Elucidating the key role of a lewis base solvent in the formation of perovskite films fabricated from the lewis adduct approach. ACS Appl. Mater. Interfaces *9*, 32868–32875.

47. Zhang, H., Li, D., Cheng, J.Q., Lin, F., Mao, J., Jen, A.K.-Y., Grätzel, M., and Choy, W.C.H. (2017). Room temperature formation of organic–inorganic lead halide perovskites: design of nanostructured and highly reactive intermediates. J. Mater. Chem. A *5*, 3599–3608.

48. Li, L., Chen, Y., Liu, Z., Chen, Q., Wang, X., and Zhou, H. (2016). The additive coordination effect on hybrids perovskite crystallization and high-performance solar cell. Adv. Mater. *28*, 9862–9868.

49. Yan, K., Long, M., Zhang, T., Wei, Z., Chen, H., Yang, S., and Xu, J. (2015). Hybrid halide perovskite solar cell precursors: colloidal chemistry and coordination engineering behind device processing for high efficiency. J. Am. Chem. Soc. *137*, 4460–4468.

50. Guo, Y., Shoyama, K., Sato, W., Matsuo, Y., Inoue, K., Harano, K., Liu, C., Tanaka, H., and Nakamura, E. (2015). Chemical pathways connecting lead(II) iodide and perovskite via polymeric plumbate(II) fiber. J. Am. Chem. Soc. *137*, 15907–15914.

51. Petrov, A.A., Sokolova, I.P., Belich, N.A., Peters, G.S., Dorovatovskii, P.V., Zubavichus, Y.V., Khrustalev, V.N., Petrov, A.V., Grätzel, M., Goodilin, E.A., and Tarasov, A.B. (2017). Crystal structure of DMF-intermediate phases uncovers the link between CH3NH3PbI3 morphology and precursor stoichiometry. J. Phys. Chem. C *121*, 20739–20743.

52. Chen, J., Xiong, Y., Rong, Y., Mei, A., Sheng, Y., Jiang, P., Hu, Y., Li, X., and Han, H. (2016). Solvent effect on the hole-conductor-free fully printable perovskite solar cells. Nano Energy *27*, 130–137.

53. Bu, T., Li, J., Li, H., Tian, C., Su, J., Tong, G., Ono, L.K., Wang, C., Lin, Z., Chai, N., et al. (2021). Lead halide-templated crystallization of methylamine-free perovskite for efficient photovoltaic modules. Science *372*, 1327–1332.

54. Wang, P., Zhang, X., Zhou, Y., Jiang, Q., Ye, Q., Chu, Z., Li, X., Yang, X., Yin, Z., and You, J. (2018). Solvent-controlled growth of inorganic perovskite films in dry environment for efficient and stable solar cells. Nat. Commun. *9*, 2225.

55. Yin, G., Zhao, H., Jiang, H., Yuan, S., Niu, T., Zhao, K., Liu, Z., and Liu, S.F. (2018). Precursor engineering for all-inorganic CsPbI2Br perovskite solar cells with 14.78% efficiency. Adv. Funct. Mater *28*, 1803269.

56. Rao, H., Ye, S., Gu, F., Zhao, Z., Liu, Z., Bian, Z., and Huang, C. (2018). Morphology controlling of all-inorganic perovskite at low temperature for efficient rigid and flexible solar cells. Adv. Energy Mater. *8*, 1800758.

57. Zai, H., Zhang, D., Li, L., Zhu, C., Ma, S., Zhao, Y., Zhao, Z., Chen, C., Zhou, H., Li, Y., and Chen, Q. (2018). Low-temperature-processed inorganic perovskite solar cells via solvent engineering with enhanced mass transport. J. Mater. Chem. A *6*, 23602–23609.

58. Zhang, S., Wu, S., Chen, W., Zhu, H., Xiong, Z., Yang, Z., Chen, C., Chen, R., Han, L., and Chen, W. (2018). Solvent engineering for efficient inverted perovskite solar cells based on inorganic CsPbI2Br light absorber. Mater. Today Energy *8*, 125–133.

59. Zhang, F., Lu, H., Tong, J., Berry, J.J., Beard, M.C., and Zhu, K. (2020). Advances in two-dimensional organic–inorganic hybrid perovskites. Energy Environ. Sci. *13*, 1154–1186.

60. Zhang, J., Zhang, L., Li, X., Zhu, X., Yu, J., and Fan, K. (2019). Binary solvent engineering for high-performance two-dimensional perovskite solar cells. ACS Sustain. Chem. Eng. *7*, 3487–3495.

61. Gao, L., Zhang, F., Xiao, C., Chen, X., Larson, B.W., Berry, J.J., and Zhu, K. (2019). Improving charge transport via intermediate-controlled crystal growth in 2D perovskite solar cells. Adv. Funct. Mater. *29*, 1901652.

62. Deng, Y., Van Brackle, C.H., Dai, X., Zhao, J., Chen, B., and Huang, J. (2019). Tailoring solvent coordination for high-speed, room-temperature blading of perovskite photovoltaic films. Sci. Adv. *5*, eaax7537.

63. Li, J., Dagar, J., Shargaieva, O., Flatken, M.A., Köbler, H., Fenske, M., Schultz, C., Stegemann, B., Just, J., Többens, D.M., et al. (2021). 20.8% Slot-die coated MAPbI3 perovskite solar cells by optimal DMSO-content and age of 2-ME based precursor inks. Adv. Energy Mater. *11*, 2003460.

64. Kim, H.-B., Choi, H., Jeong, J., Kim, S., Walker, B., Song, S., and Kim, J.Y. (2014). Mixed solvents for the optimization of morphology in solution-processed, inverted-type perovskite/fullerene hybrid solar cells. Nanoscale *6*, 6679–6683.

65. Wang, X., Ran, X., Liu, X., Gu, H., Zuo, S., Hui, W., Lu, H., Sun, B., Gao, X., Zhang, J., et al. (2020). Tailoring component interaction for air-processed efficient and stable all-inorganic perovskite photovoltaic. Angew. Chem. Int. Ed. Engl. *59*, 13354–13361.

66. Qiu, J., Xia, Y., Chen, Y., and Huang, W. (2019). Management of crystallization kinetics for efficient and stable low-dimensional Ruddlesden-Popper (LDRP) lead-free perovskite solar cells. Adv. Sci. *6*, 1800793.

67. Zhou, Z., Wang, Z., Zhou, Y., Pang, S., Wang, D., Xu, H., Liu, Z., Padture, N.P., and Cui, G. (2015). Methylamine-gas-induced defect-healing behavior of CH3NH3PbI3 thin films for perovskite solar cells. Angew. Chem. Int. Ed. Engl. *54*, 9705–9709.

68. Huang, X., Chen, R., Deng, G., Han, F., Ruan, P., Cheng, F., Yin, J., Wu, B., and Zheng, N. (2020). Methylamine-dimer-induced phase transition toward MAPbI3 films and high-efficiency perovskite solar modules. J. Am. Chem. Soc. *142*, 6149–6157.

69. Zong, Y., Zhou, Y., Ju, M., Garces, H.F., Krause, A.R., Ji, F., Cui, G., Zeng, X.C., Padture, N.P., and Pang, S. (2016). Thin-film transformation of NH4PbI3 to CH3NH3PbI3 perovskite: a methylamine-induced conversion-healing process. Angew. Chem. Int. Ed. *55*, 14723–14727.

70. Pang, S., Zhou, Y., Wang, Z., Yang, M., Krause, A.R., Zhou, Z., Zhu, K., Padture, N.P., and Cui, G. (2016). Transformative evolution of organolead triiodide perovskite thin films from strong room-temperature solid-gas interaction between HPbI3-CH3NH2 precursor pair. J. Am. Chem. Soc. *138*, 750–753.

71. Li, C., Pang, S., Xu, H., and Cui, G. (2017). Methylamine gas based synthesis and healing process toward upscaling of perovskite solar cells: progress and perspective. Sol. RRL *1*, 1700076.

72. Chen, H., Ye, F., Tang, W., He, J., Yin, M., Wang, Y., Xie, F., Bi, E., Yang, X., Grätzel, M., and Han, L. (2017). A solvent- and vacuum-free route to large-area perovskite films for efficient solar modules. Nature *550*, 92–95.

73. Fan, H., Li, F., Wang, P., Gu, Z., Huang, J.H., Jiang, K.J., Guan, B., Yang, L.M., Zhou, X., and Song, Y. (2020). Methylamine-assisted growth of uniaxial-oriented perovskite thin films with millimeter-sized grains. Nat. Commun. *11*, 5402.

74. Noel, N.K., Habisreutinger, S.N., Wenger, B., Klug, M.T., Hörantner, M.T., Johnston, M.B., Nicholas, R.J., Moore, D.T., and Snaith, H.J.

(2017). A low viscosity, low boiling point, clean solvent system for the rapid crystallisation of highly specular perovskite films. Energy Environ. Sci. *10*, 145–152.

75. Wang, K., Wu, C., Hou, Y., Yang, D., Li, W., Deng, G., Jiang, Y., and Priya, S. (2020). A nonionic and low-entropic MA(MMA)nPbI3-ink for fast crystallization of perovskite thin films. Joule *4*, 615–630.

76. Bai, Y., Xiao, S., Hu, C., Zhang, T., Meng, X., Li, Q., Yang, Y., Wong, K.S., Chen, H., and Yang, S. (2017). A pure and stable intermediate phase is key to growing aligned and vertically monolithic perovskite crystals for efficient PIN planar perovskite solar cells with high processibility and stability. Nano Energy *34*, 58–68.

77. Hu, C., Shivarudraiah, S.B., Sung, H.H.Y., Williams, I.D., Halpert, J.E., and Yang, S. (2021). Discovery of a new intermediate enables one-step deposition of high-quality perovskite films via solvent engineering. Sol. RRL *5*, 2000712.

78. Xiao, M.D., Huang, F.Z., Huang, W.C., Dkhissi, Y., Zhu, Y., Etheridge, J., Gray-Weale, A., Bach, U., Cheng, Y.B., and Spiccia, L. (2014). A fast deposition-crystallization procedure for highly efficient lead iodide perovskite thin-film solar cells. Angew. Chem. Int. Ed. Engl. *53*, 9898–9903.

79. Ahn, N., Son, D.Y., Jang, I.H., Kang, S.M., Choi, M., and Park, N.G. (2015). Highly reproducible perovskite solar cells with average efficiency of 18.3% and best efficiency of 19.7% fabricated via lewis base adduct of lead(II) iodide. J. Am. Chem. Soc. *137*, 8696–8699.

80. Douglas, T.B. (1948). Vapor pressure of methyl sulfoxide from 20 to 50°. Calculation of the heat of vaporization. J. Am. Chem. Soc. *70*, 2001–2002.

81. Zhou, Y., Yang, M., Wu, W., Vasiliev, A.L., Zhu, K., and Padture, N.P. (2015). Room-temperature crystallization of hybrid-perovskite thin films via solvent–solvent extraction for high-performance solar cells. J. Mater. Chem. A *3*, 8178–8184.

82. Lee, J.W., Dai, Z., Lee, C., Lee, H.M., Han, T.H., De Marco, N., Lin, O., Choi, C.S., Dunn, B., Koh, J., et al. (2018). Tuning molecular interactions for highly reproducible and efficient formamidinium perovskite solar cells via adduct approach. J. Am. Chem. Soc. *140*, 6317–6324.

83. Wang, S., Chen, Y., Li, R., Xu, Y., Feng, J., Yang, D., Yuan, N., Zhang, W.H., Liu, S.F., and Ding, J. (2020). Superior textured film and process tolerance enabled by intermediate-state engineering for high-efficiency perovskite solar cells. Adv. Sci. *7*, 1903009.

84. Feng, J., Zhu, X., Yang, Z., Zhang, X., Niu, J., Wang, Z., Zuo, S., Priya, S., Liu, S.F., and Yang, D. (2018). Record efficiency stable flexible perovskite solar cell using effective additive assistant strategy. Adv. Mater. *30*, e1801418.

85. Li, Y., Wang, J., Yuan, Y., Dong, X., and Wang, P. (2017). Anti-solvent dependent device performance in CH3NH3PbI3 solar cells: the role of intermediate phase content in the as-prepared thin films. Sustain. Energy Fuels *1*, 1041–1048.

86. Moot, T., Marshall, A.R., Wheeler, L.M., Habisreutinger, S.N., Schloemer, T.H., Boyd, C.C., Dikova, D.R., Pach, G.F., Hazarika, A., McGehee, M.D., et al. (2020). CsI-antisolvent adduct formation in all-inorganic metal halide perovskites. Adv. Energy Mater. *10*, 1903365.

87. Dahlman, C.J., DeCrescent, R.A., Venkatesan, N.R., Kennard, R.M., Wu, G., Everest, M.A., Schuller, J.A., and Chabinyc, M.L. (2019). Controlling solvate intermediate growth for phase-pure organic lead iodide Ruddlesden–Popper (C4H9NH3)2(CH3NH3)n−1PbnI3n+1 perovskite thin films. Chem. Mater. *31*, 5832–5844.

88. Mateen, M., Arain, Z., Liu, X., Liu, C., Yang, Y., Ding, Y., Ma, S., Ren, Y., Wu, Y., Tao, Y., et al. (2020). High-performance mixed-cation mixed-halide perovskite solar cells enabled by a facile intermediate engineering technique. J. Power Sources *448*, 227386.

89. Zhao, Y., and Zhu, K. (2014). Solution chemistry engineering toward high-efficiency perovskite solar cells. J. Phys. Chem. Lett. *5*, 4175–4186.

90. Wang, D., Liu, Z., Zhou, Z., Zhu, H., Zhou, Y., Huang, C., Wang, Z., Xu, H., Jin, Y., Fan, B., et al. (2014). Reproducible one-step fabrication of compact MAPbI3–xClx thin films derived from mixed-lead-halide precursors. Chem. Mater. *26*, 7145–7150.

91. Zhao, Y.X., and Zhu, K. (2014). Efficient planar perovskite solar cells based on 1.8 eV band gap CH3NH3PbI2Br nanosheets via thermal decomposition. J. Am. Chem. Soc. *136*, 12241–12244.

92. Yu, H., Wang, F., Xie, F., Li, W., Chen, J., and Zhao, N. (2014). The role of chlorine in the formation process of "CH3NH3PbI3-xClx" perovskite. Adv. Funct. Mater. *24*, 7102–7108.

93. Zhang, W., Saliba, M., Moore, D.T., Pathak, S.K., Hörantner, M.T., Stergiopoulos, T., Stranks, S.D., Eperon, G.E., Alexander-Webber, J.A., Abate, A., et al. (2015). Ultrasmooth organic-inorganic perovskite thin-film formation and crystallization for efficient planar heterojunction solar cells. Nat. Commun. *6*, 6142.

94. Liu, D., Li, C., Zhang, C., Wang, Z., Zhang, H., Tian, J., and Pang, S. (2017). Blended additive manipulated morphology and crystallinity transformation toward high performance perovskite solar cells. RSC Adv *7*, 51944–51949.

95. Ke, W., Xiao, C., Wang, C., Saparov, B., Duan, H.S., Zhao, D., Xiao, Z., Schulz, P., Harvey, S.P., Liao, W., et al. (2016). Employing lead thiocyanate additive to reduce the hysteresis and boost the fill factor of planar perovskite solar cells. Adv. Mater. *28*, 5214–5221.

96. Zuo, C., and Ding, L. (2014). An 80.11% FF record achieved for perovskite solar cells by using the NH4Cl additive. Nanoscale *6*, 9935–9938.

97. Si, H., Liao, Q., Kang, Z., Ou, Y., Meng, J., Liu, Y., Zhang, Z., and Zhang, Y. (2017). Deciphering the NH4PbI3 intermediate phase for simultaneous improvement on nucleation and crystal growth of perovskite. Adv. Funct. Mater. *27*, 1701804.

98. Yang, Y., Liu, C., Syzgantseva, O.A., Syzgantseva, M.A., Ma, S., Ding, Y., Cai, M., Liu, X., Dai, S., and Nazeeruddin, M.K. (2021). Defect suppression in oriented 2D perovskite solar cells with efficiency over 18% via rerouting crystallization pathway. Adv. Energy Mater. *11*, 2002966.

99. Duan, B., Ren, Y., Xu, Y., Chen, W., Ye, Q., Huang, Y., Zhu, J., and Dai, S. (2017). Identification and characterization of a new intermediate to obtain high quality perovskite films with hydrogen halides as additives. Inorg. Chem. Front. *4*, 473–480.

100. Li, G., Zhang, T., and Zhao, Y. (2015). Hydrochloric acid accelerated formation of planar CH3NH3PbI3 perovskite with high humidity tolerance. J. Mater. Chem. A *3*, 19674–19678.

101. Heo, J.H., Song, D.H., and Im, S.H. (2014). Planar CH3NH3PbBr3 hybrid solar cells with 10.4% power conversion efficiency, fabricated by controlled crystallization in the spin-coating process. Adv. Mater. *26*, 8179–8183.

102. Heo, J.H., Song, D.H., Han, H.J., Kim, S.Y., Kim, J.H., Kim, D., Shin, H.W., Ahn, T.K., Wolf, C., Lee, T.-W., et al. (2015). Planar CH3NH3PbI3 perovskite solar cells with constant 17.2% average power conversion efficiency irrespective of the scan rate. Adv. Mater. *27*, 3424–3430.

103. Lu, H., Liu, Y., Ahlawat, P., Mishra, A., Tress, W.R., Eickemeyer, F.T., Yang, Y., Fu, F., Wang, Z., Avalos, C.E., et al. (2020). Vapor-assisted deposition of highly efficient, stable black-phase FAPbI3 perovskite solar cells. Science *370*, eabb8985.

104. Meng, H., Shao, Z., Wang, L., Li, Z., Liu, R., Fan, Y., Cui, G., and Pang, S. (2020). Chemical composition and phase evolution in DMAI-derived inorganic perovskite solar cells. ACS Energy Lett *5*, 263–270.

105. Steele, J.A., Jin, H., Dovgaliuk, I., Berger, R.F., Braeckevelt, T., Yuan, H., Martin, C., Solano, E., Lejaeghere, K., Rogge, S.M.J., et al. (2019). Thermal unequilibrium of strained black CsPbI3 thin films. Science *365*, 679–684.

106. Eperon, G.E., Paternò, G.M., Sutton, R.J., Zampetti, A., Haghighirad, A.A., Cacialli, F., and Snaith, H.J. (2015). Inorganic caesium lead iodide perovskite solar cells. J. Mater. Chem. A *3*, 19688–19695.

107. Zhang, T., Dar, M.I., Li, G., Xu, F., Guo, N., Grätzel, M., and Zhao, Y. (2017). Bication lead iodide 2D perovskite component to stabilize inorganic α-CsPbI3 perovskite phase for high-efficiency solar cells. Sci. Adv. *3*, e1700841.

108. Ye, Q., Zhao, Y., Mu, S., Ma, F., Gao, F., Chu, Z., Yin, Z., Gao, P., Zhang, X., and You, J. (2019). Cesium lead inorganic solar cell with efficiency beyond 18% via reduced charge recombination. Adv. Mater. *31*, e1905143.

109. Zheng, Y., Yang, X., Su, R., Wu, P., Gong, Q., and Zhu, R. (2020). High-performance CsPbIxBr3-x all-inorganic perovskite solar cells with efficiency over 18% via spontaneous interfacial manipulation. Adv. Funct. Mater. *30*, 2000457.

110. Ke, W., Spanopoulos, I., Stoumpos, C.C., and Kanatzidis, M.G. (2018). Myths and reality of HPbI3 in halide perovskite solar cells. Nat. Commun. *9*, 4785.

111. Bing, J., Lee, D.S., Zheng, J., Zhang, M., Li, Y., Kim, J., Lau, C.F.J., Cho, Y., Green, M.A., Huang, S., et al. (2019). Deconstruction-assisted perovskite formation for sequential solution processing of Cs0.15(MA0.7FA0.3) 0.85PbI3 solar cells. Sol. Energy Mater. Sol. Cells *203*, 110200.

112. Bian, H., Wang, H., Li, Z., Zhou, F., Xu, Y., Zhang, H., Wang, Q., Ding, L., Liu, S.F., and Jin, Z. (2020). Unveiling the effects of hydrolysis-derived DMAI/DMAPbIx intermediate compound on the performance of CsPbI3 solar cells. Adv. Sci. *7*, 1902868.

113. Pei, Y., Liu, Y., Li, F., Bai, S., Jian, X., and Liu, M. (2019). Unveiling property of hydrolysis-derived DMAPbI3 for perovskite devices: composition engineering, defect mitigation, and stability optimization. iScience *15*, 165–172.

114. Sun, X., Shao, Z., Rao, Y., Meng, H., Gao, C., Chen, C., Liu, D., Lv, P., Li, Z., Wang, X., et al. (2021). A low-temperature additive-involved leaching method for highly efficient inorganic perovskite solar cells. Adv. Energy Mater. *11*, 2002754.

115. Shao, Z., Meng, H., Du, X., Sun, X., Lv, P., Gao, C., Rao, Y., Chen, C., Li, Z., Wang, X., et al. (2020). Cs4PbI6-mediated synthesis of thermodynamically stable FA0.15Cs0.85PbI3 perovskite solar cells. Adv. Mater. *32*, e2001054.

116. Lau, C.F.J., Wang, Z., Sakai, N., Zheng, J., Liao, C.H., Green, M., Huang, S., Snaith, H.J., and Ho-Baillie, A. (2019). Fabrication of efficient and stable CsPbI3 perovskite solar cells through cation exchange process. Adv. Energy Mater. *9*, 1901685.

117. Zhang, T., Wang, Y., Wang, X., Wu, M., Liu, W., and Zhao, Y. (2019). Organic salt mediated growth of phase pure and stable all-inorganic CsPbX3 (X=I, Br) perovskites for efficient photovoltaics. Sci. Bull. *64*, 1773–1779.

118. Jiang, Q., Zhao, Y., Zhang, X., Yang, X., Chen, Y., Chu, Z., Ye, Q., Li, X., Yin, Z., and You, J. (2019). Surface passivation of perovskite film for efficient solar cells. Nat. Photonics *13*, 460–466.

119. Li, N., Luo, Y., Chen, Z., Niu, X., Zhang, X., Lu, J., Kumar, R., Jiang, J., Liu, H., Guo, X., et al. (2020). Microscopic degradation in formamidinium-cesium lead iodide perovskite solar cells under operational stressors. Joule *4*, 1743–1758.

120. Yi, C., Luo, J., Meloni, S., Boziki, A., Ashari-Astani, N., Grätzel, C., Zakeeruddin, S.M., Röthlisberger, U., and Grätzel, M. (2016). Entropic stabilization of mixed A-cation ABX3 metal halide perovskites for high performance perovskite solar cells. Energy Environ. Sci. *9*, 656–662.

121. Zhang, Y., Zhou, Z., Ji, F., Li, Z., Cui, G., Gao, P., Oveisi, E., Nazeeruddin, M.K., and Pang, S. (2018). Trash into treasure: δ-FAPbI3 polymorph stabilized MAPbI3 perovskite with power conversion efficiency beyond 21. Adv. Mater. *30*, e1707143.

122. Kim, G., Min, H., Lee, K.S., Lee, D.Y., Yoon, S.M., and Seok, S.I. (2020). Impact of strain relaxation on performance of α-formamidinium lead iodide perovskite solar cells. Science *370*, 108–112.